

# Building and evaluation of a PBPK model for clarithromycin in healthy adults

|                                                    |                                                                                                                                                                                       |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version                                            | 2.0-OSP12.3                                                                                                                                                                           |
| based on <i>Model Snapshot and Evaluation Plan</i> | <a href="https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/releases/tag/v2.0">https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/releases/tag/v2.0</a> |
| OSP Version                                        | 12.3                                                                                                                                                                                  |
| Qualification Framework Version                    | 3.5                                                                                                                                                                                   |

This evaluation report and the corresponding PK-Sim project file are filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/>

# Table of Contents

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- 1 Introduction
- 2 Methods
  - 2.1 Modeling Strategy
  - 2.2 Data
  - 2.3 Model Parameters and Assumptions
- 3 Results and Discussion
  - 3.1 Final input parameters
  - 3.2 Diagnostics Plots
  - 3.3 Concentration-Time Profiles
    - 3.3.1 Model Building
    - 3.3.2 Model Verification
- 4 Conclusion
- 5 References

# 1 Introduction

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Clarithromycin is a widely prescribed macrolide antibiotic and a substrate and mechanism-based inactivator of CYP3A4. Furthermore, clarithromycin is a substrate and inhibitor of P-gp and an inhibitor of OATP1B1 and OATP1B3 ([Eberl 2007](#), [Seithel 2007](#)). Clarithromycin has been proposed as one of the best alternative CYP3A4 inhibitors for clinical DDI studies to avoid further use of ketoconazole.

Objectives were to develop a fully mechanistic PBPK model for clarithromycin, describing its metabolism by CYP3A4 and its mechanism-based inactivation of the respective enzyme as well as its inhibition of P-gp.

The presented clarithromycin model was developed by Moj et al. ([Moj 2017](#)) and revised by Hanke et al. ([Hanke 2018](#)).

## 2 Methods

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### 2.1 Modeling Strategy

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The general workflow for building an adult PBPK model has been described by Kuepfer et al. ([Kuepfer 2016](#)). Relevant information on the anthropometry (height, weight) was gathered from the respective clinical study, if reported. Information on physiological parameters (e.g. blood flows, organ volumes, hematocrit) in adults was gathered from the literature and has been incorporated in PK-Sim® as described previously ([Willmann 2007](#)). The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available 'PK-Sim® Ontogeny Database Version 7.3' ([PK-Sim Ontogeny Database Version 7.3](#)).

A typical European individual was used for the development of the clarithromycin model. The relative tissue-specific expression of CYP3A4 was implemented in accordance with literature information using the PK-Sim expression database RT-PCR profile. Enterohepatic recirculation was enabled as it is active under physiological conditions.

Unknown parameters (see [Section 2.3.4](#)) were identified using the Parameter Identification module provided in PK-Sim®.

The model was then verified by simulating the PK of additional clinical studies including a dose range of 100 to 1200 mg administered as single dose or as multiple doses.

Details about input data (physicochemical, *in vitro* and clinical) can be found in [Section 2.2](#).

Details about the structural model and its parameters can be found in [Section 2.3](#).

## 2.2 Data

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### 2.2.1 In vitro / physicochemical Data

A literature search was performed to collect available information on physiochemical properties of clarithromycin. The obtained information from literature is summarized in the table below.

| Parameter                 | Unit   | Value                    | Source                         | Description                                                                                             |
|---------------------------|--------|--------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------|
| MW                        | g/mol  | 747.95                   | <a href="#">drugbank.ca</a>    | Molecular weight                                                                                        |
| pK <sub>a</sub> (base)    |        | 8.99                     | <a href="#">McFarland 1997</a> | Acid dissociation constant                                                                              |
| Solubility (pH)           | mg/L   | 12170 (2.4)              | <a href="#">Salem 2003</a>     | Solubility                                                                                              |
| logP                      |        | 2.3                      | <a href="#">Lappin 2011</a>    | Partition coefficient between octanol and water                                                         |
| fu                        | %      | 28.0                     | <a href="#">Davey 1991</a>     | Fraction unbound in plasma                                                                              |
|                           | %      | 30.0                     | <a href="#">Chu 1993b</a>      | Fraction unbound in plasma                                                                              |
|                           | %      | 40.0                     | <a href="#">Noreddin 2002</a>  | Fraction unbound in plasma                                                                              |
| CYP3A4 K <sub>m</sub>     | μmol/L | 48.7                     | <a href="#">Rodrigues 1997</a> | Michaelis-Menten constant for CYP3A4 metabolism                                                         |
| CL <sub>ren</sub>         | L/h    | 6.66 - 12.8 <sup>a</sup> | <a href="#">Rodvold 1999</a>   | Renal clearance                                                                                         |
| CYP3A4 K <sub>i</sub>     | μmol/L | 2.25                     | <a href="#">Polasek 2006</a>   | Conc. for half-maximal inactivation measured in recombinant CYP3A4                                      |
|                           | μmol/L | 29.5                     | <a href="#">Polasek 2006</a>   | Conc. for half-maximal inactivation measured in human liver microsomes                                  |
|                           | μmol/L | 41.4                     | <a href="#">Ito 2003</a>       | Conc. for half-maximal inactivation measured in human liver microsomes for α-hydroxylation of midazolam |
|                           | μmol/L | 37.0                     | <a href="#">Ito 2003</a>       | Conc. for half-maximal inactivation measured in human liver microsomes for 4-hydroxylation of midazolam |
|                           | μmol/L | 5.49                     | <a href="#">Mayhew 2000</a>    | Conc. for half-maximal inactivation measured in human liver microsomes                                  |
| CYP3A4 k <sub>inact</sub> | 1/min  | 0.04                     | <a href="#">Polasek 2006</a>   | Maximum inactivation rate measured in recombinant CYP3A4                                                |
|                           | 1/min  | 0.05                     | <a href="#">Polasek 2006</a>   | Maximum inactivation rate measured in human liver microsomes                                            |
|                           | 1/min  | 0.0423                   | <a href="#">Ito 2003</a>       | Maximum inactivation rate measured in human liver microsomes for α-hydroxylation of midazolam           |
|                           | 1/min  | 0.0459                   | <a href="#">Ito 2003</a>       | Maximum inactivation rate measured in human liver microsomes for 4-hydroxylation of midazolam           |

| Parameter                   | Unit              | Value           | Source                       | Description                                                  |
|-----------------------------|-------------------|-----------------|------------------------------|--------------------------------------------------------------|
|                             | 1/min             | 0.072           | <a href="#">Mayhew 2000</a>  | Maximum inactivation rate measured in human liver microsomes |
| P-gp $K_i$                  | $\mu\text{mol/L}$ | 4.1             | <a href="#">Eberl 2007</a>   | Conc. for half-maximal inhibition                            |
| OATP1B1<br>$\text{IC}_{50}$ | $\mu\text{mol/L}$ | $5.3 \pm 1.3^b$ | <a href="#">Vermeer 2016</a> | Half-maximal inhibitory concentration                        |
| OATP1B3<br>$\text{IC}_{50}$ | $\mu\text{mol/L}$ | $14 \pm 2^b$    | <a href="#">Vermeer 2016</a> | Half-maximal inhibitory concentration                        |

<sup>a</sup> denotes range of reported values

<sup>b</sup> denotes mean  $\pm$  standard error of the mean of the measurements (two assays, each performed in triplicate)

## 2.2.2 Clinical Data

A literature search was performed to collect available clinical data on clarithromycin in healthy adults. The clarithromycin model was developed using 17 clinical studies covering a dosing range from 100 to 1200 mg.

### 2.2.2.1 Model Building

The following studies were used for model building (training data):

| Publication               | Arm / Treatment / Information used for model building                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">Chu 1992b</a> | Healthy subjects with intravenous infusion of 250 mg over 45 min                                    |
| <a href="#">Chu 1993a</a> | Healthy subjects with oral administration of 250 or 500 mg as single dose or twice daily for 5 days |

### 2.2.2.2 Model Verification

The following studies were used for model verification (test data):

| Publication                        | Arm / Treatment / Information used for model building                                 |
|------------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">Chu 1992a</a>          | Healthy Subjects with oral administration of single doses ranging from 100 to 1200 mg |
| <a href="#">Kees 1995</a>          | Healthy subjects with oral administration of 250 or 500 mg as single or multiple dose |
| <a href="#">Rengelshausen 2003</a> | Oral administration of 250 mg twice a day for 1.5 days                                |
| <a href="#">Abduljalil 2009</a>    | Oral administration of 500 mg twice a day for 3.5 days                                |

## 2.3 Model Parameters and Assumptions

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### 2.3.1 Absorption

The specific intestinal permeability was optimized during parameter identification to accurately describe the absorption of clarithromycin after oral administration.

### 2.3.2 Distribution

Values for lipophilicity and fraction unbound in plasma were fixed to literature values (namely to 2.3 (Lappin 2011) and 0.30 (Chu 1993b) for lipophilicity and fraction unbound, respectively).

It was not possible to adequately describe the observed plasma concentration-time profile after intravenous administration using standard input parameters (e.g. lipophilicity) and calculation methods. Simulated concentration-time profiles overestimated  $C_{\max}$  and underestimated the observed data for time to  $C_{\max}$  ( $T_{\max}$ ). According to literature data, clarithromycin accumulates in mononuclear and polymorphonuclear leukocytes, probably via active transport (Ishiguro 1989). Implementing this process in the model improved the model fit significantly. Due to limited knowledge on this active transport, this process was technically implemented in the model by adjusting the permeability of clarithromycin across the membrane of the red blood cells ( $P_{\text{(blood cells} \rightarrow \text{plasma)}}$  and  $P_{\text{(plasma} \rightarrow \text{blood cells)}}$ ).

After testing the available organ-plasma partition coefficient and cell permeability calculation methods built in PK-Sim, observed clinical data was best described by choosing the partition coefficient calculation by Rodgers and Rowland and cellular permeability calculation by PK-Sim Standard.

### 2.3.3 Metabolism and Elimination

Metabolism was described using Michaelis-Menten kinetics, while the Michaelis-Menten constant  $K_m$  was taken from in-vitro experiments from literature and the turnover rate  $k_{\text{cat}}$  was optimized during parameter identification.

$K_i$  and  $k_{\text{inact}}$  to describe the mechanism-based inhibition of CYP3A4 were optimized during parameter identification.

A kidney plasma clearance was implemented to describe the renal elimination of clarithromycin. The specific renal clearance was optimized during parameter identification.

### 2.3.4 Automated Parameter Identification

This is the result of the final parameter identification.

| Model Parameter                                            | Optimized Value | Unit              |
|------------------------------------------------------------|-----------------|-------------------|
| <code>kcat</code> (CYP3A4)                                 | 76.5            | 1/min             |
| <code>Specific_clearance</code> in process renal clearance | 0.87            | 1/min             |
| <code>Specific_intestinal_permeability</code>              | 1.23 E-6        | dm/min            |
| <code>P</code> (plasma->blood cells)                       | 3.62 E-5        | dm/min            |
| <code>P</code> (blood cells->plasma)                       | 1.04 E-6        | dm/min            |
| <code>K_kinact_half</code> ( $K_I$ )                       | 6.04            | $\mu\text{mol/L}$ |
| <code>kinact</code> ( $k_{\text{inact}}$ )                 | 0.04            | 1/min             |



## 3 Results and Discussion

The PBPK model for clarithromycin was developed and verified with clinical pharmacokinetic data.

The model was evaluated covering data from studies including in particular

- intravenous and oral administrations
- a dose range of 100 mg to 1200 mg
- single and multiple doses

The model quantifies metabolism via CYP3A4, including also the mechanism-based inhibition of the respective enzyme, as well as elimination via kidney. The model also includes inhibition of P-gp.

The next sections show:

1. the final model parameters for the building blocks: [Section 3.1](#).
2. the overall goodness of fit: [Section 3.2](#).
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: [Section 3.3](#).

### 3.1 Final input parameters

The compound parameter values of the final PBPK model are illustrated below.

#### Compound: Clarithromycin

##### Parameters

| Name                                             | Value           | Value Origin                       | Alternative | Default |
|--------------------------------------------------|-----------------|------------------------------------|-------------|---------|
| Solubility at reference pH                       | 12.17 mg/ml     | Publication-Salem 2003             | Measurement | True    |
| Reference pH                                     | 2.4             | Publication-Salem 2003             | Measurement | True    |
| Lipophilicity                                    | 2.3 Log Units   | Publication-Lappin 2011            | Measurement | True    |
| Fraction unbound (plasma, reference value)       | 0.3             | Publication-Chu 1993               | Measurement | True    |
| Specific intestinal permeability (transcellular) | 1.23E-06 dm/min | Parameter Identification-optimized | fit         | True    |
| Is small molecule                                | Yes             |                                    |             |         |
| Molecular weight                                 | 747.9534 g/mol  |                                    |             |         |
| Plasma protein binding partner                   | Albumin         |                                    |             |         |

## Calculation methods

| Name                    | Value               |
|-------------------------|---------------------|
| Partition coefficients  | Rodgers and Rowland |
| Cellular permeabilities | PK-Sim Standard     |

## Processes

### Metabolizing Enzyme: CYP3A4-fit

Molecule: CYP3A4

#### Parameters

| Name                               | Value                      | Value Origin               |
|------------------------------------|----------------------------|----------------------------|
| In vitro Vmax for liver microsomes | 0 pmol/min/mg mic. protein |                            |
| Km                                 | 48.7 $\mu$ mol/l           | Publication-Rodrigues 1997 |
| kcat                               | 76.5 1/min                 | Parameter Identification   |

### Systemic Process: Renal Clearances-fitted

Species: Human

#### Parameters

| Name                          | Value          | Value Origin |
|-------------------------------|----------------|--------------|
| Body weight                   | 71.5 kg        | Unknown      |
| Blood flow rate (kidney)      | 1.31 l/min     | Unknown      |
| Fraction unbound (experiment) | 0.4            |              |
| Plasma clearance              | 1.75 ml/min/kg |              |

### Inhibition: P-gp-Eberl (2007)

Molecule: P-gp

#### Parameters

| Name | Value      | Value Origin           |
|------|------------|------------------------|
| Ki   | 4.1 µmol/l | Publication-Eberl 2007 |

### Inhibition: CYP3A4-fitted

Molecule: CYP3A4

#### Parameters

| Name          | Value       | Value Origin |
|---------------|-------------|--------------|
| kinact        | 0.04 1/min  |              |
| K_kinact_half | 6.04 µmol/l |              |

### Inhibition: OATP1B1-Vermeer 2016

Molecule: OATP1B1

#### Parameters

| Name | Value      | Value Origin             |
|------|------------|--------------------------|
| Ki   | 5.3 µmol/l | Publication-Vermeer 2016 |

### Inhibition: OATP1B3-Vermeer 2016

Molecule: OATP1B3

#### Parameters

| Name | Value     | Value Origin             |
|------|-----------|--------------------------|
| Ki   | 14 µmol/l | Publication-Vermeer 2016 |

## Formulation: Tablet Clarithromycin

Type: Weibull

#### Parameters

| Name                             | Value | Value Origin |
|----------------------------------|-------|--------------|
| Dissolution time (50% dissolved) | 5 min |              |
| Lag time                         | 0 min |              |
| Dissolution shape                | 2.9   |              |
| Use as suspension                | No    |              |

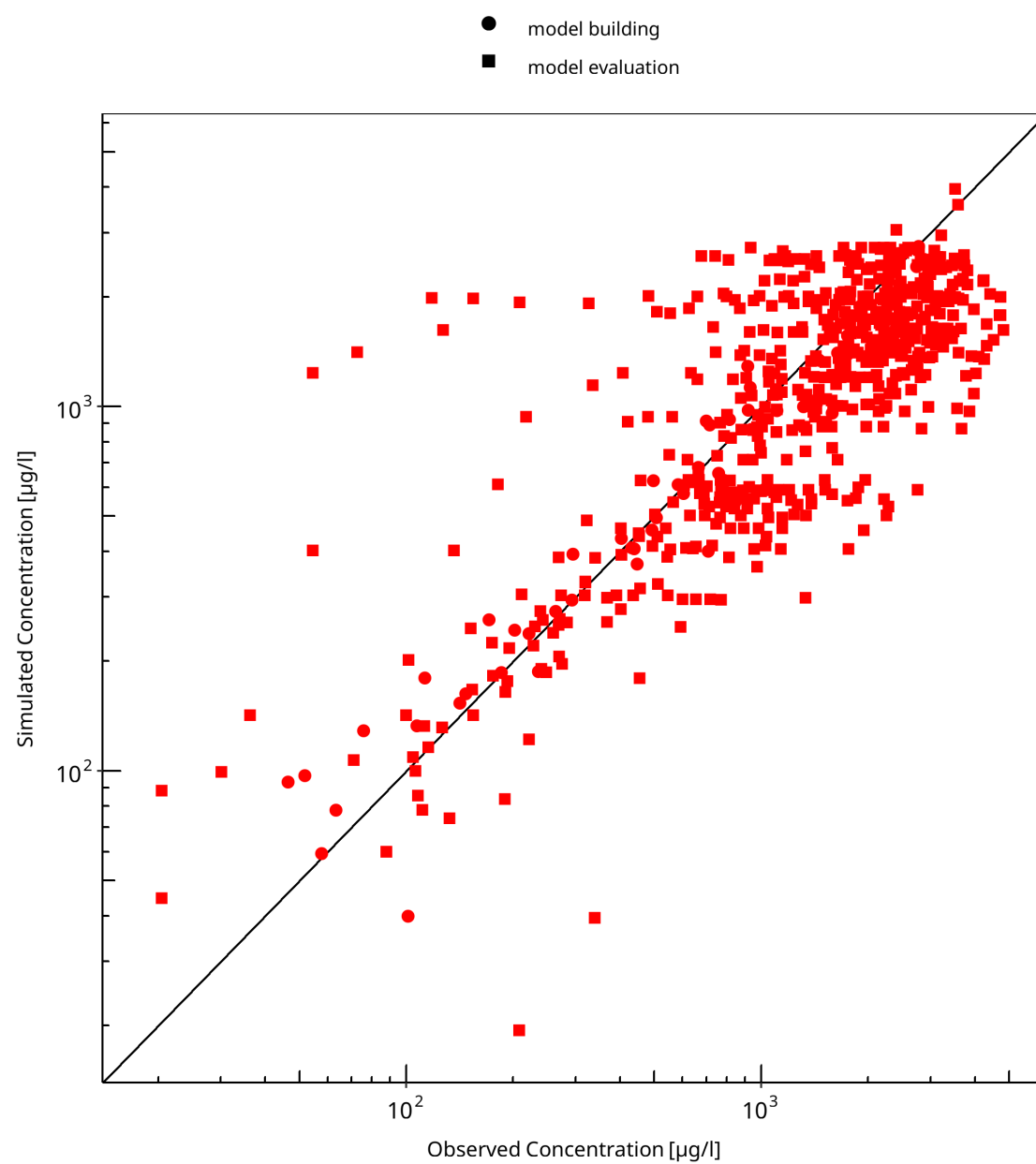
## 3.2 Diagnostics Plots

Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data used presented in [Section 2.2.2](#).

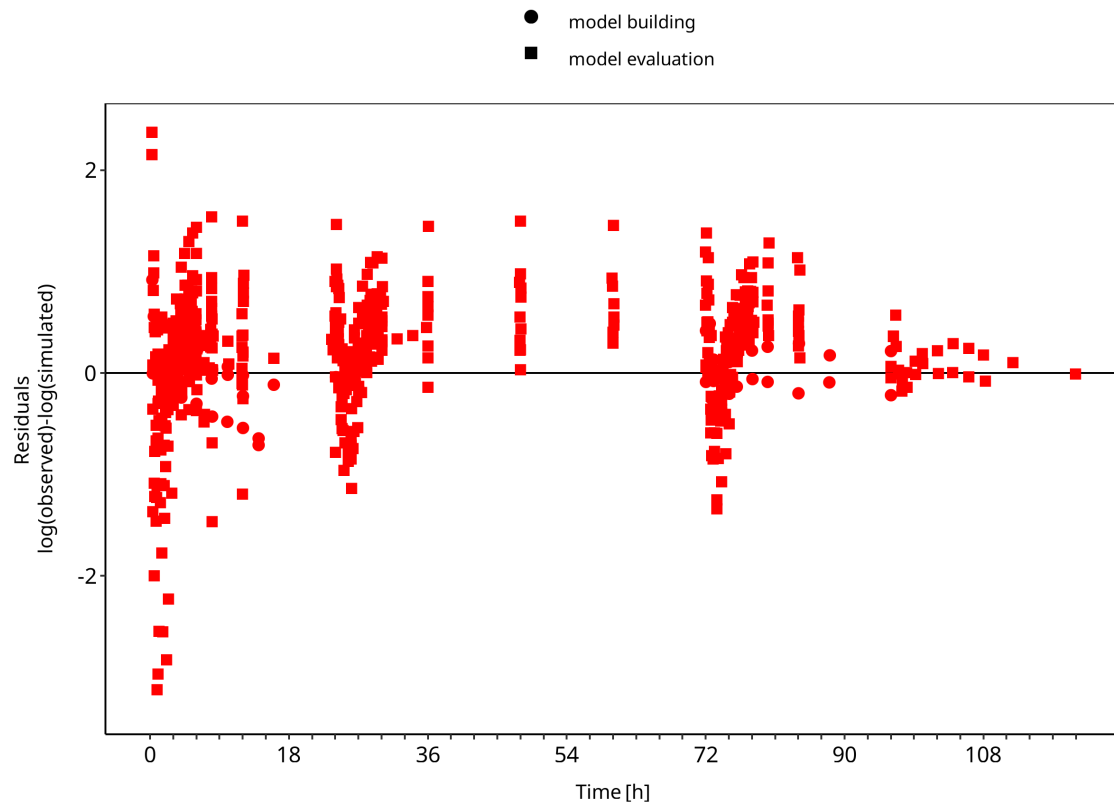
The first plot shows simulated versus observed plasma concentrations, the second weighted residuals versus time.

**Table 3-1: GMFE for Goodness of fit plot for concentration in plasma**

| Group            | GMFE |
|------------------|------|
| model building   | 1.21 |
| model evaluation | 1.62 |
| All              | 1.58 |



**Figure 3-1: Goodness of fit plot for concentration in plasma**

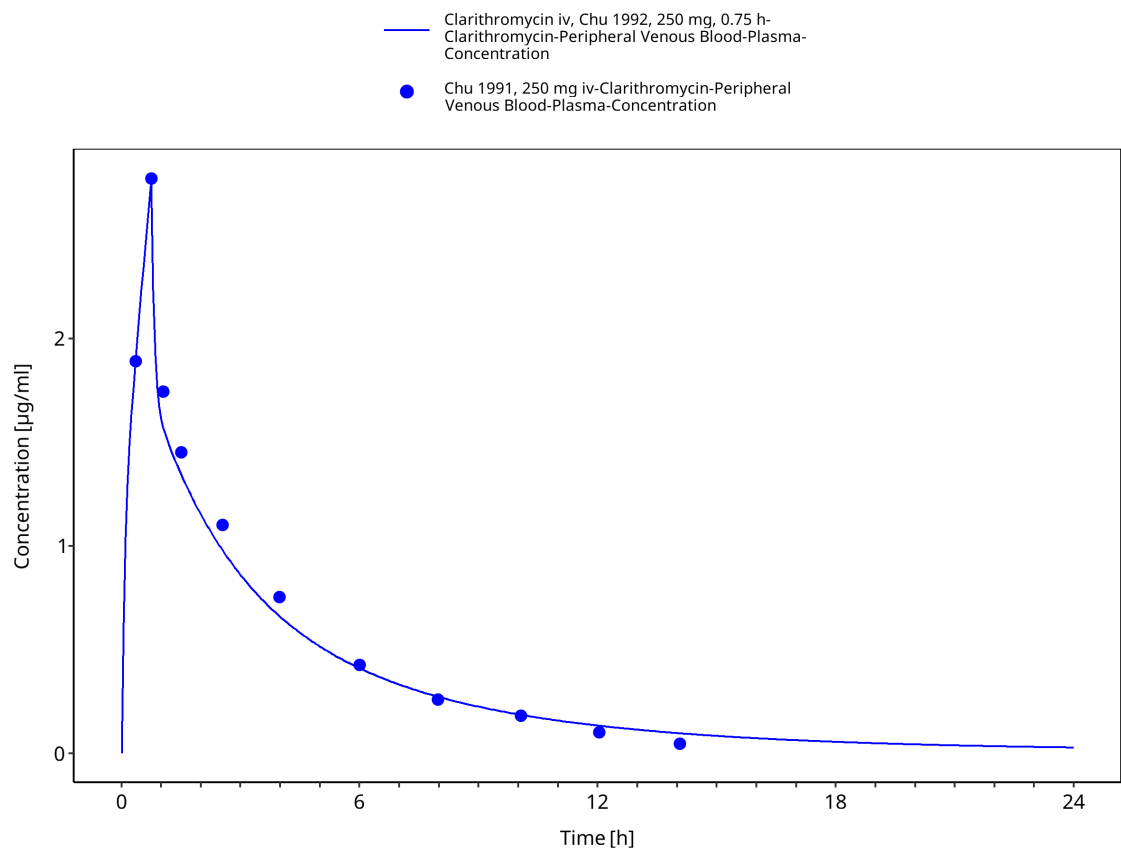


**Figure 3-2: Goodness of fit plot for concentration in plasma**

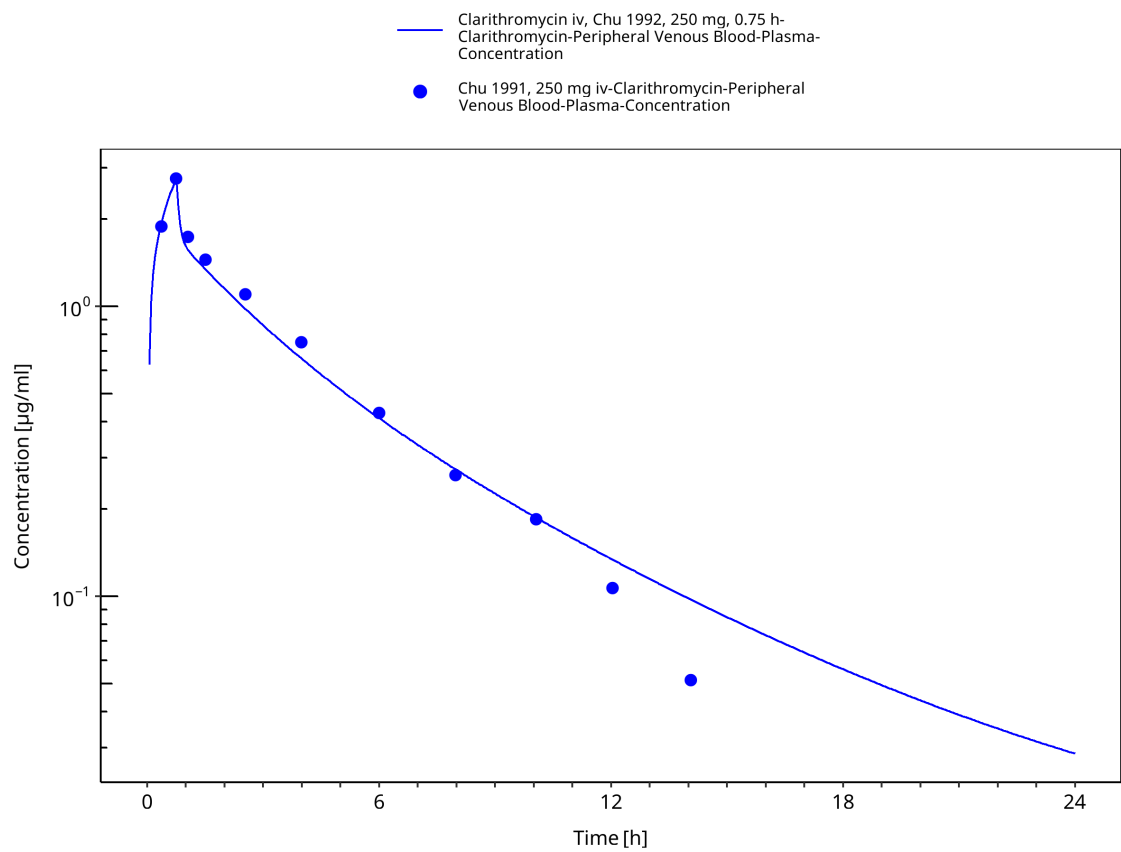
## 3.3 Concentration-Time Profiles

Simulated versus observed concentration-time profiles of all data listed in [Section 2.2.2](#) are presented below.

### 3.3.1 Model Building

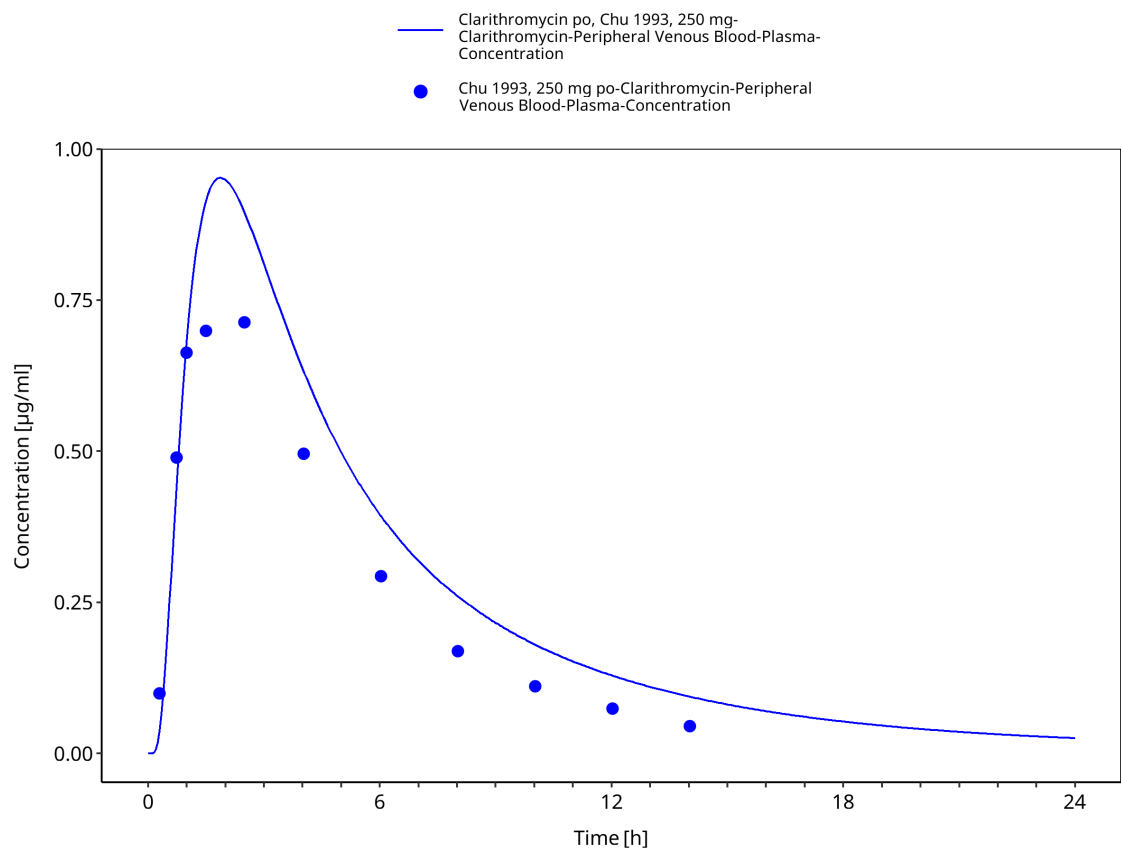


**Figure 3-3: Time Profile Analysis**

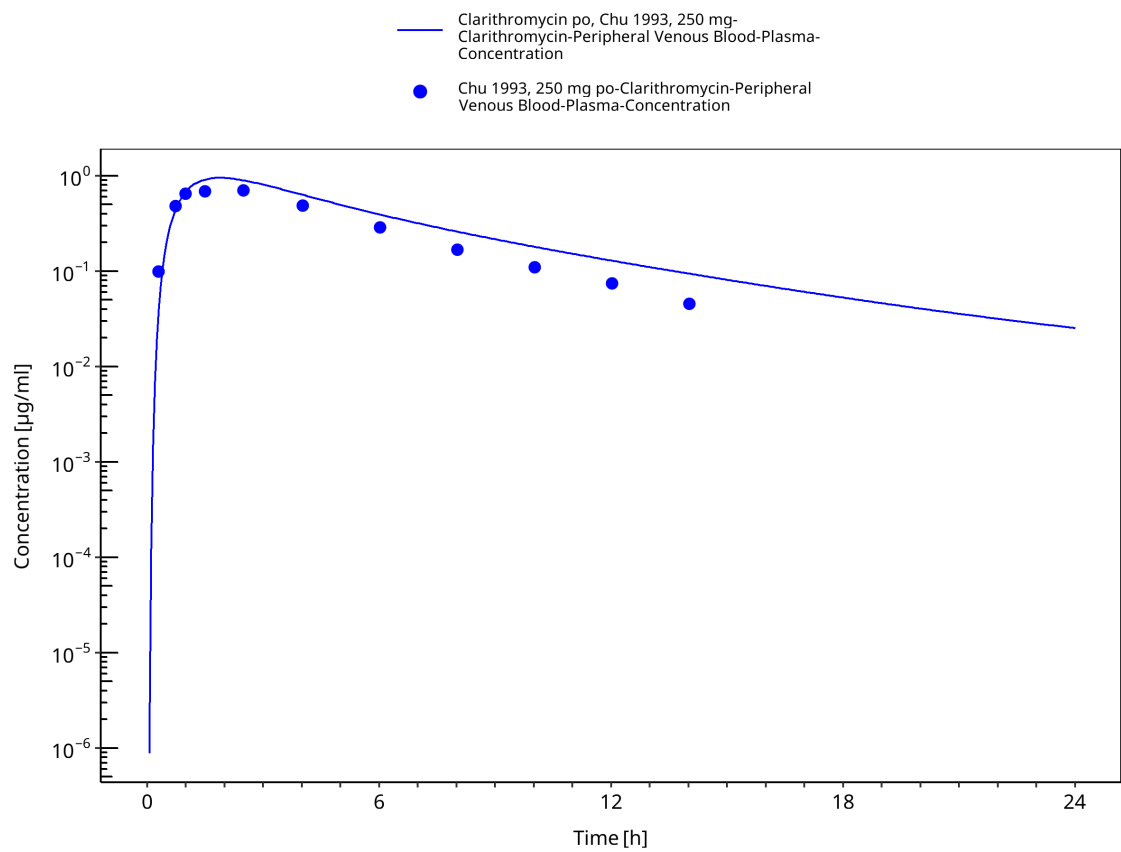


**Figure 3-4: Time Profile Analysis 1**





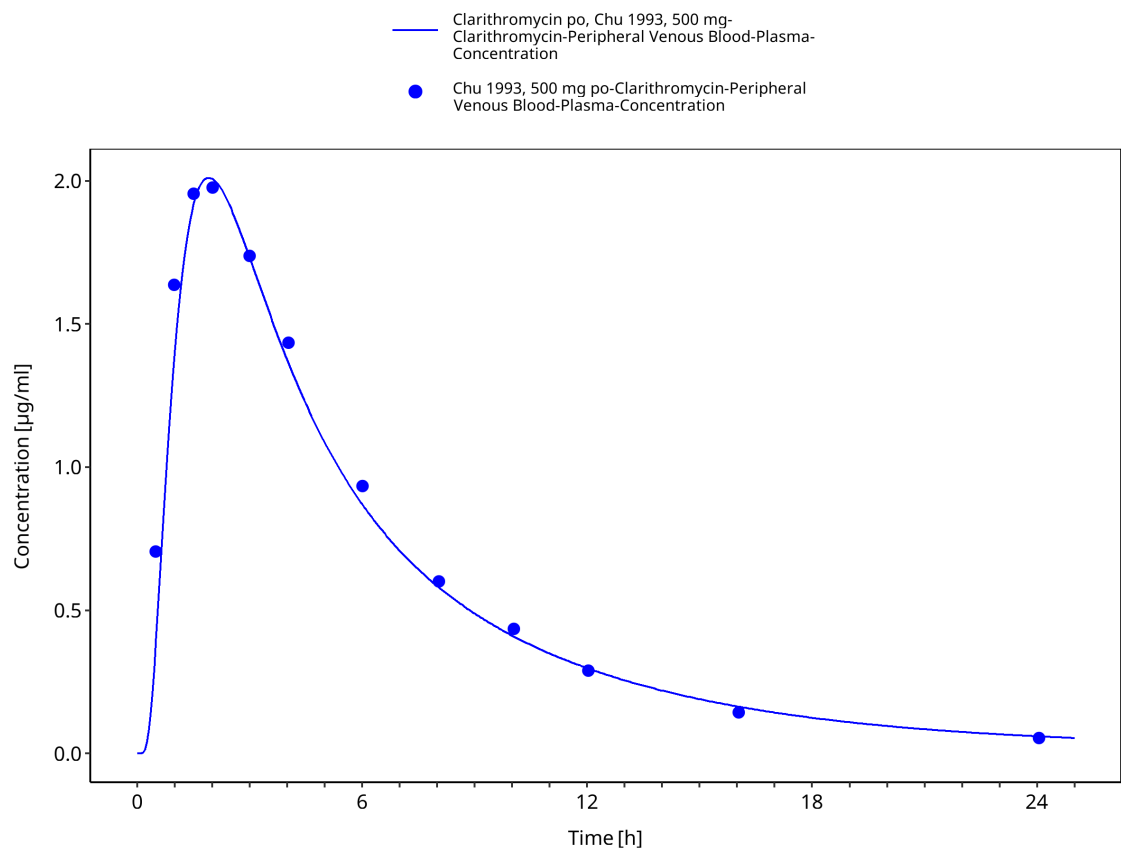
**Figure 3-5: Time Profile Analysis**



**Figure 3-6: Time Profile Analysis 1**







**Figure 3-9: Time Profile Analysis**

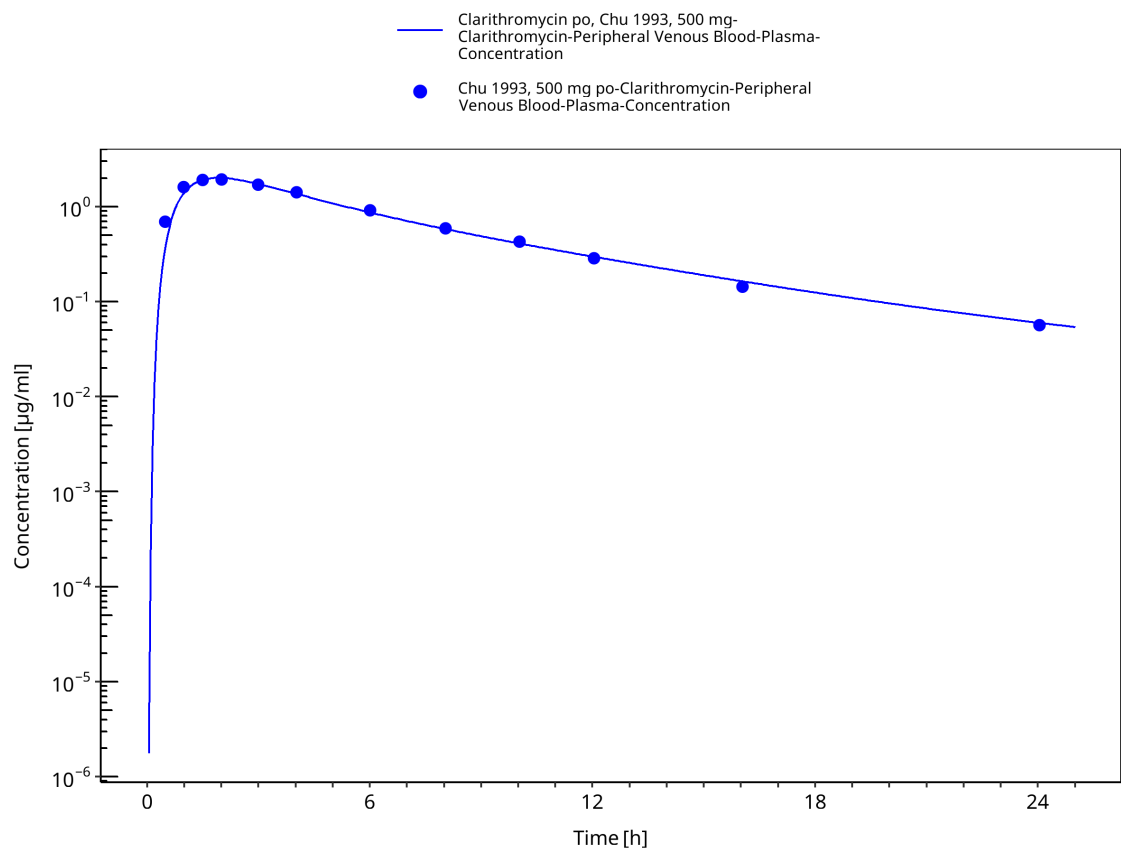
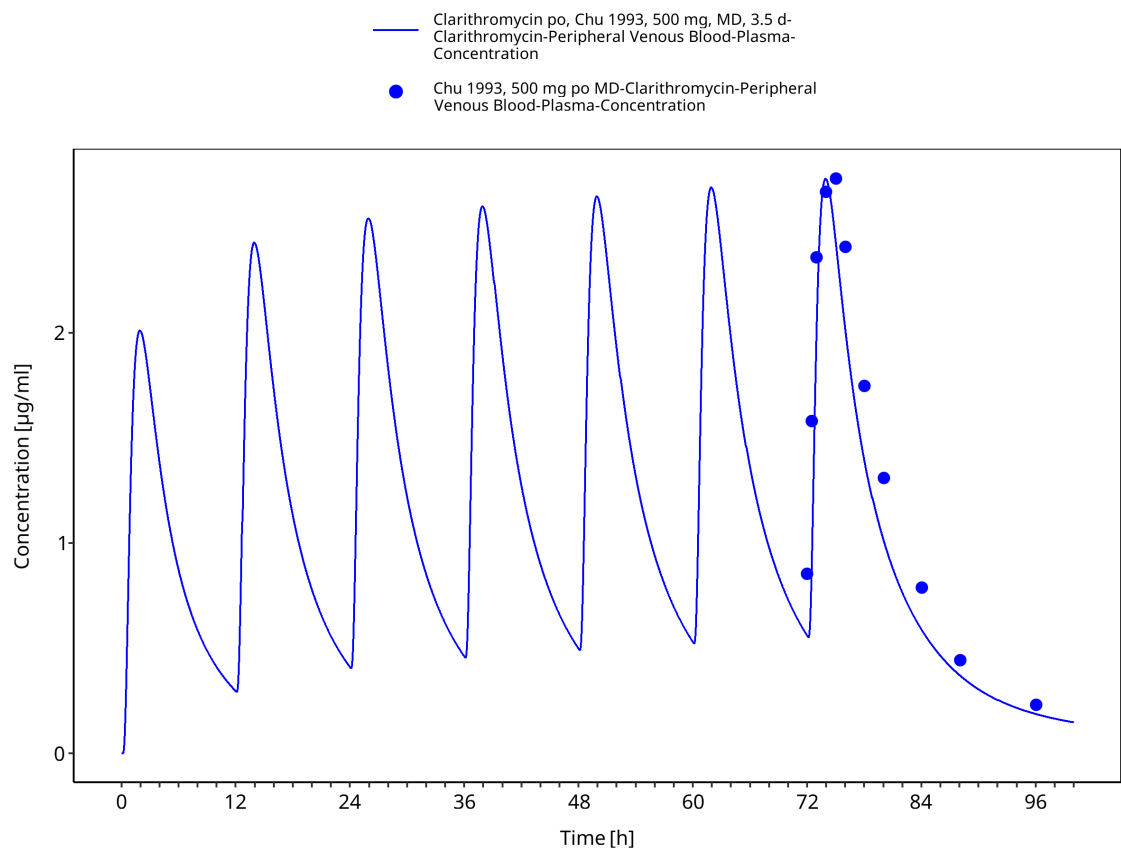


Figure 3-10: Time Profile Analysis 1



**Figure 3-11: Time Profile Analysis**

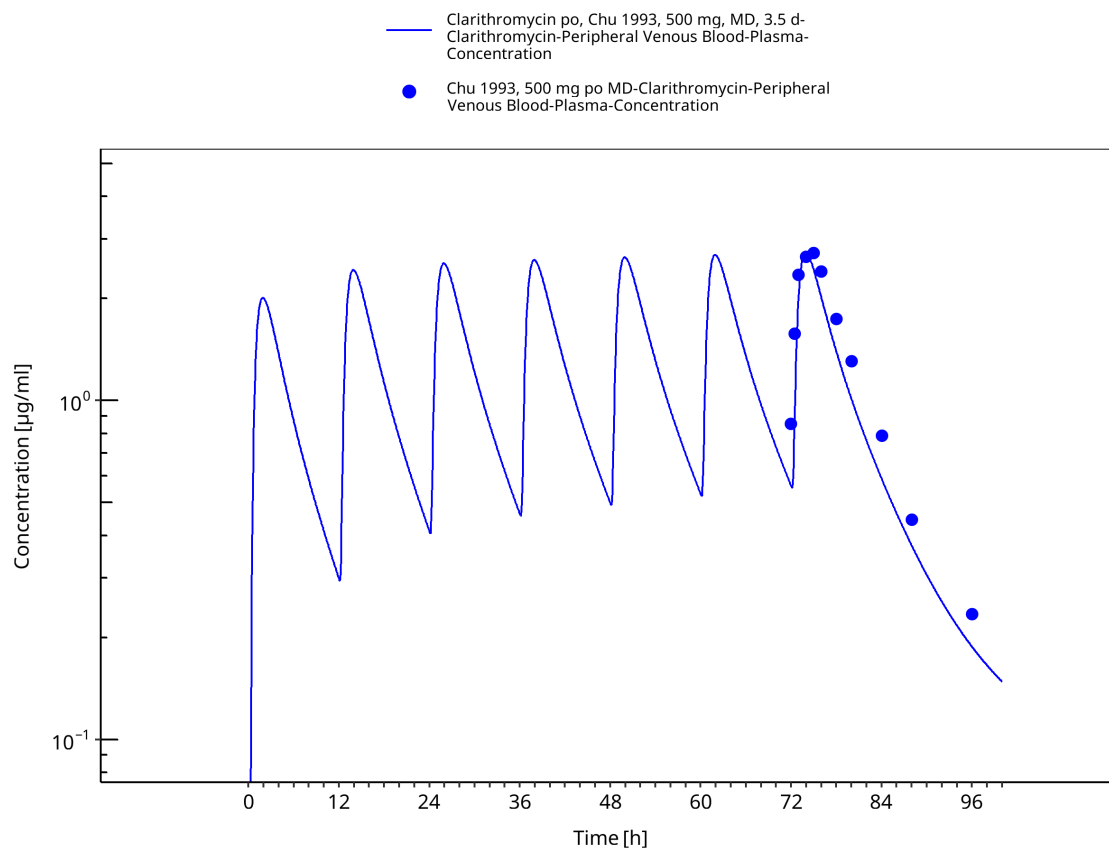
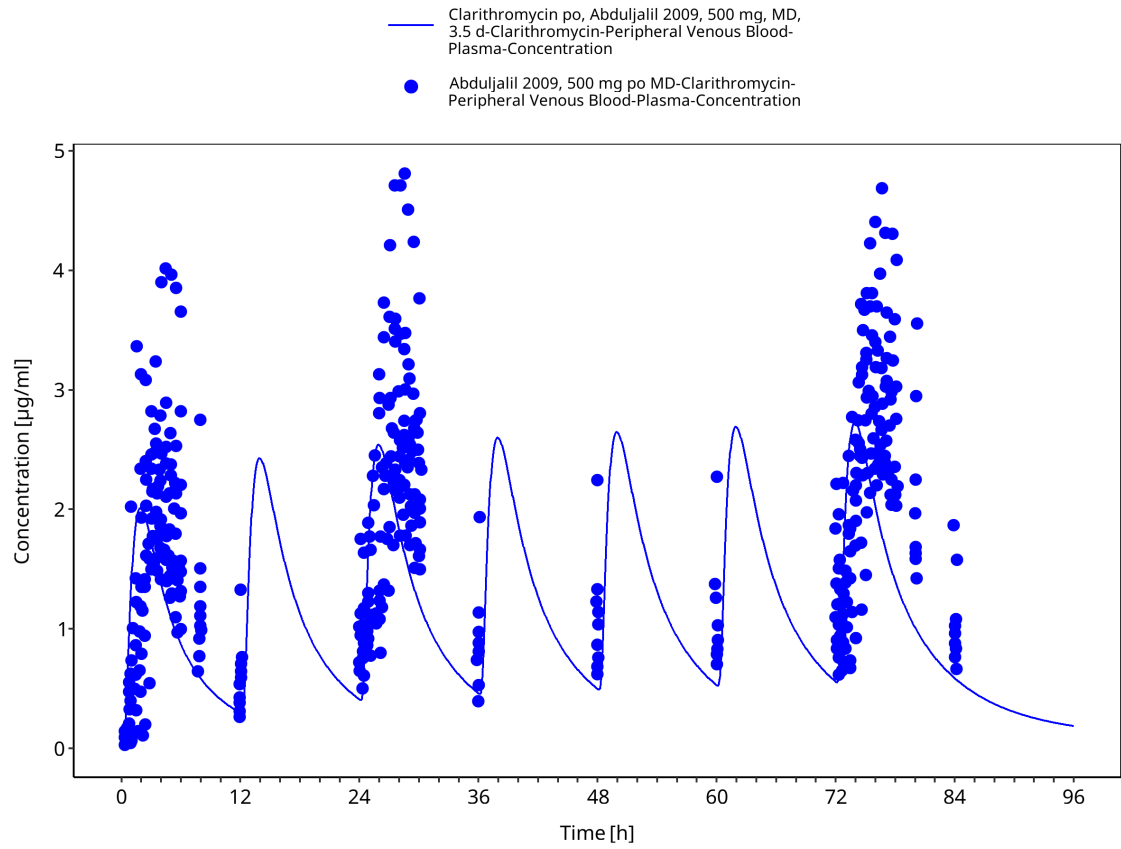


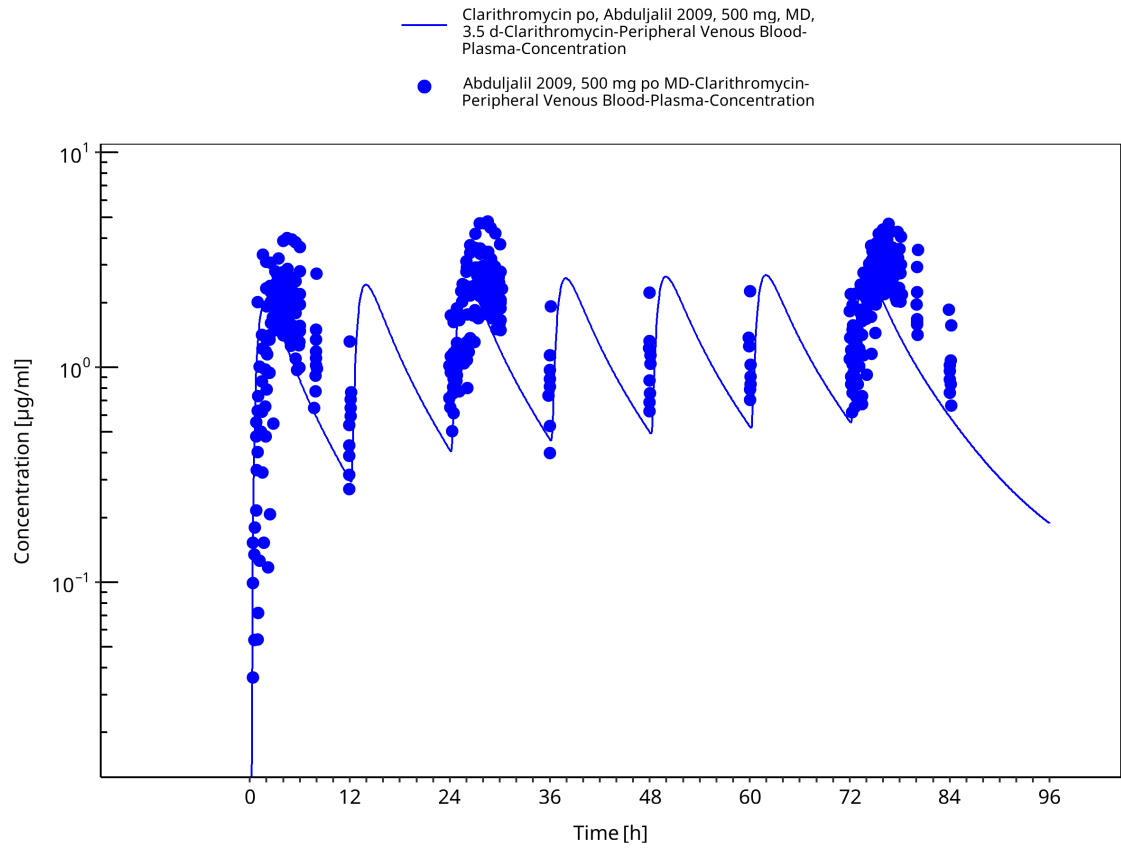
Figure 3-12: Time Profile Analysis 1

### 3.3.2 Model Verification

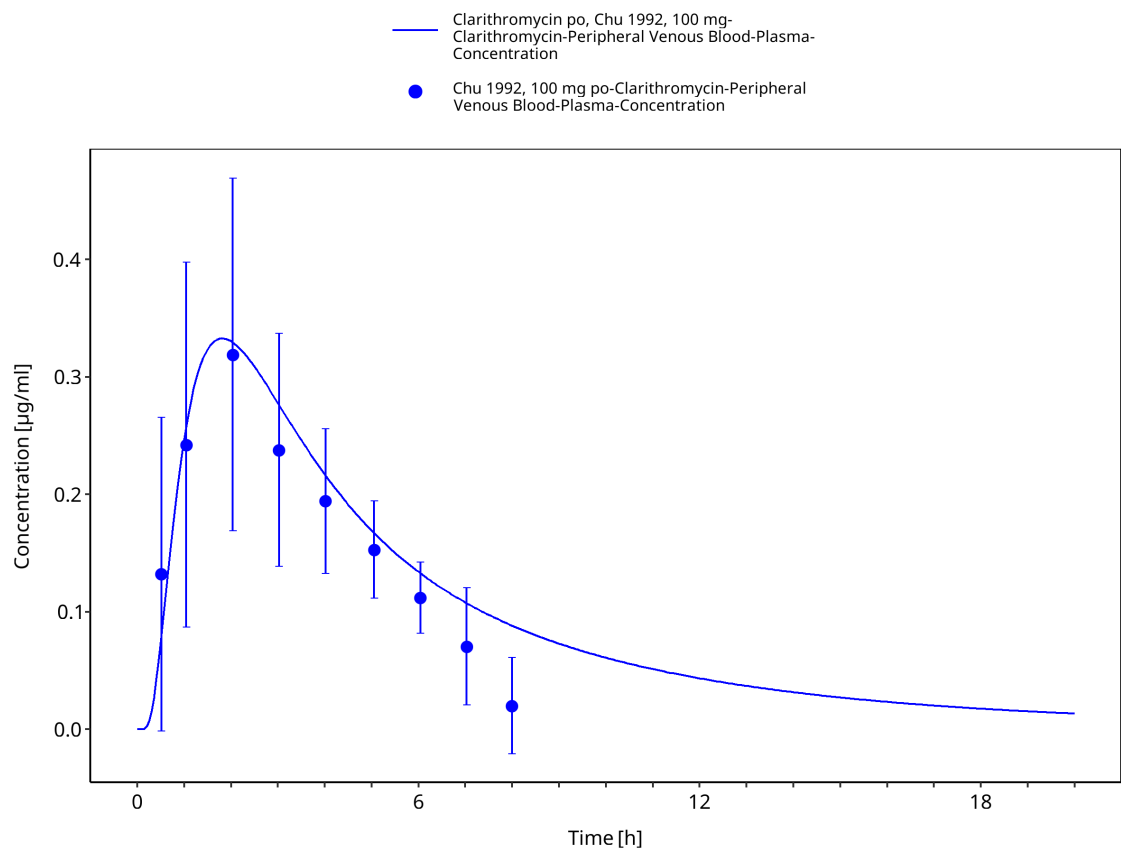




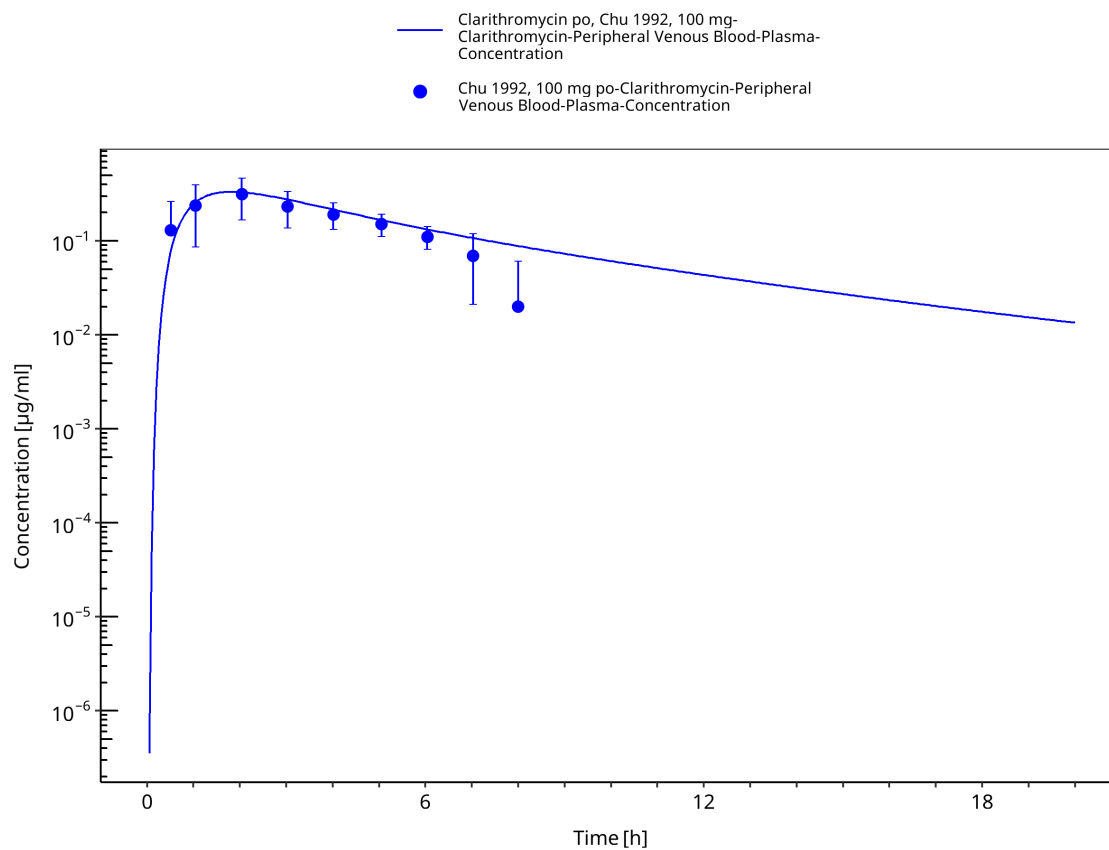
**Figure 3-13: Time Profile Analysis**



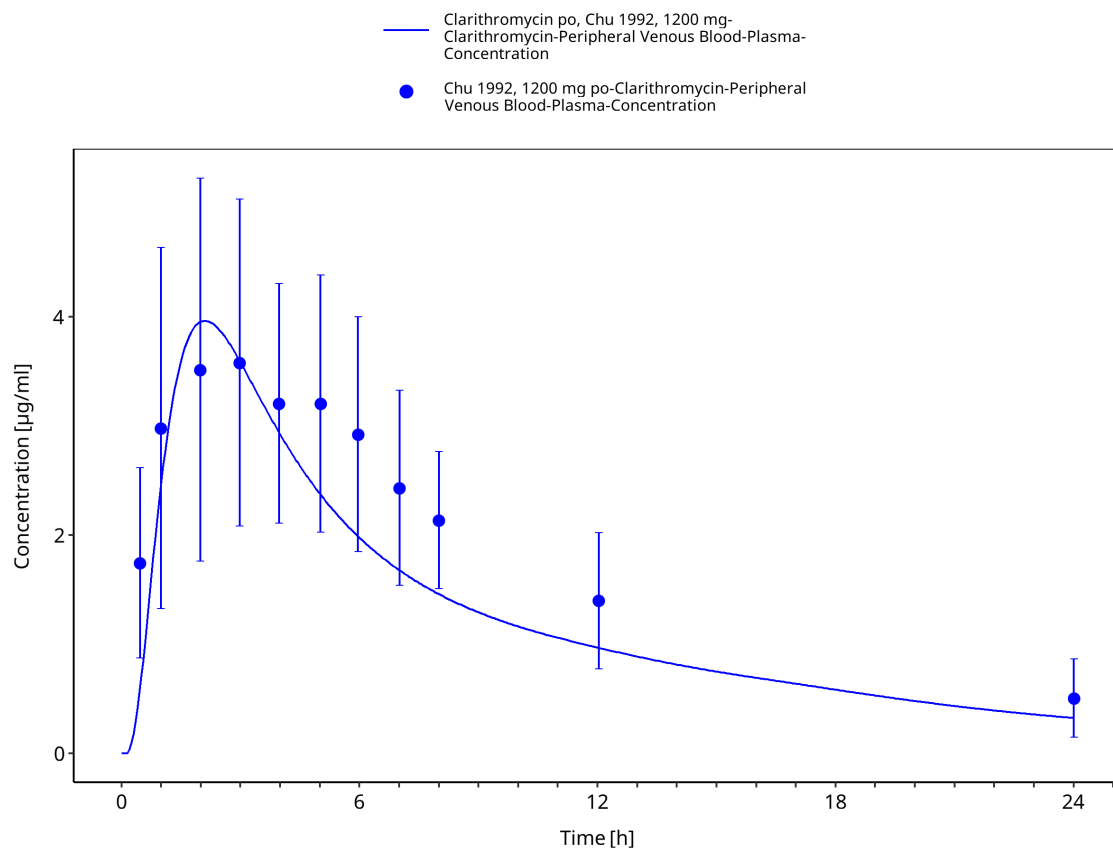
**Figure 3-14: Time Profile Analysis 1**



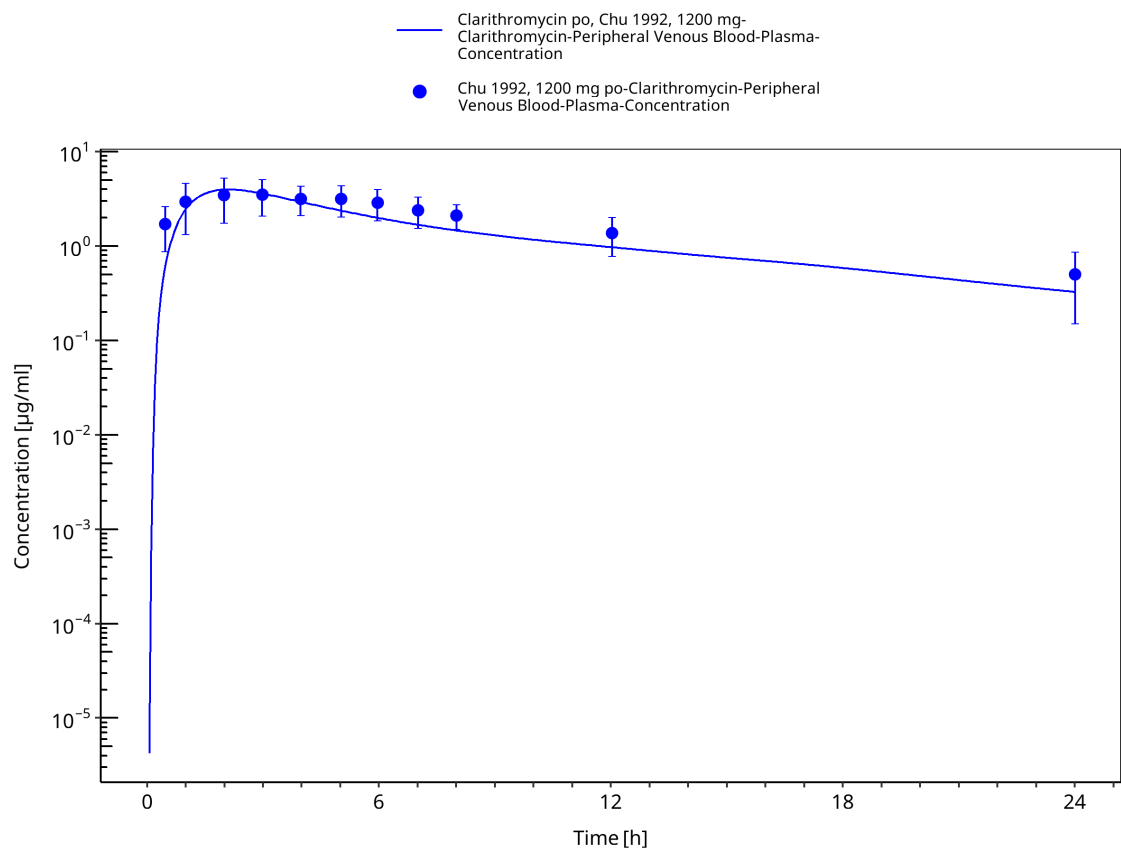
**Figure 3-15: Time Profile Analysis**



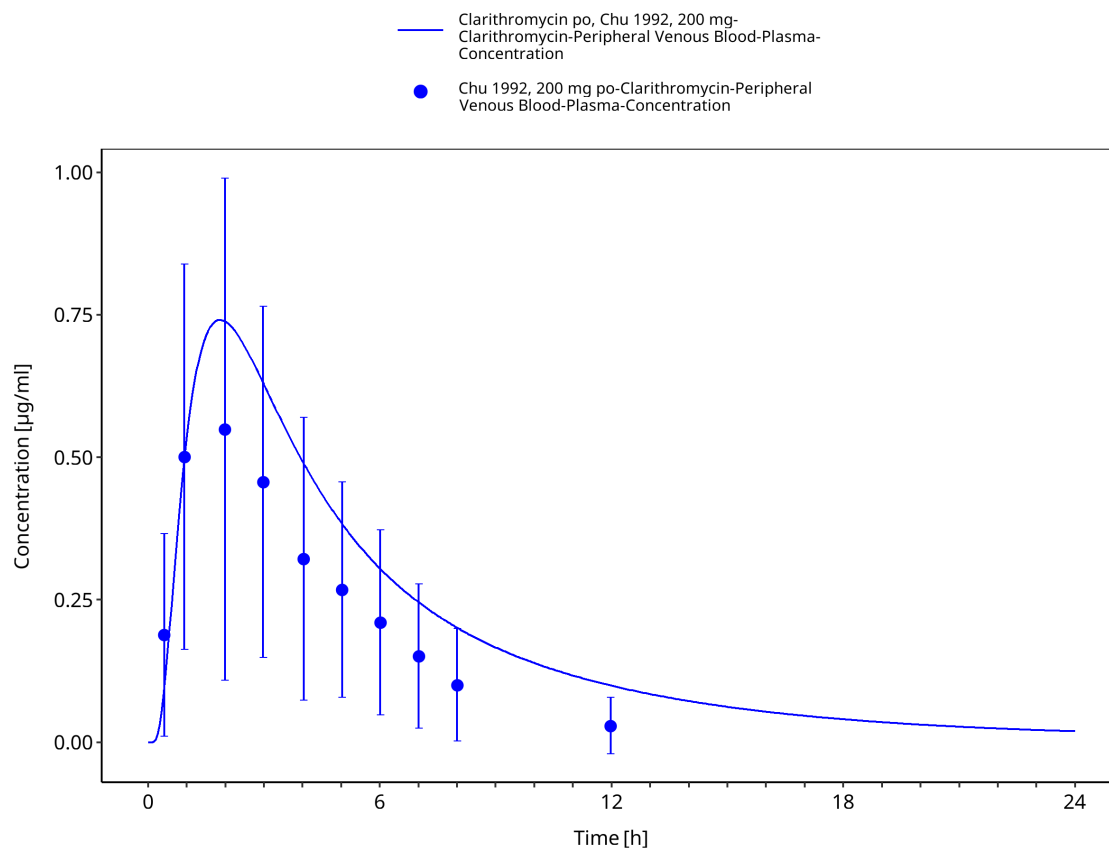
**Figure 3-16: Time Profile Analysis 1**



**Figure 3-17: Time Profile Analysis**



**Figure 3-18: Time Profile Analysis 1**



**Figure 3-19: Time Profile Analysis**

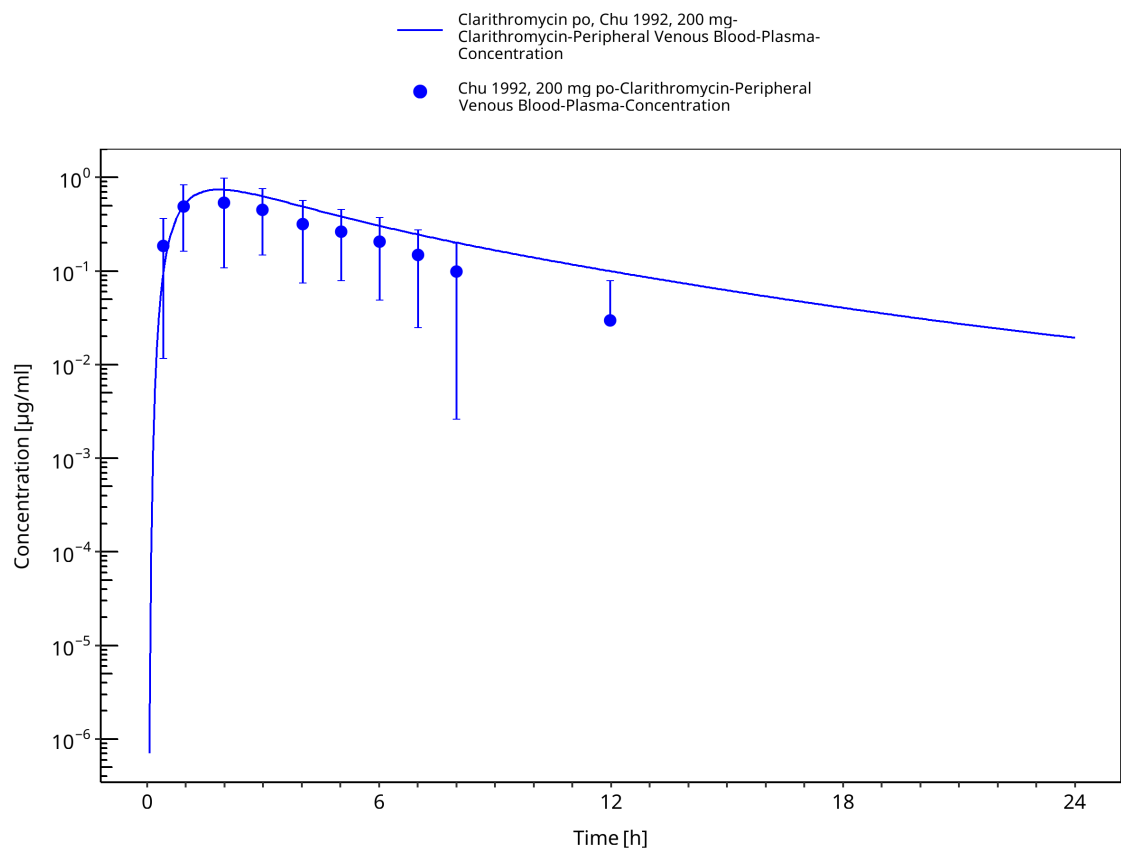
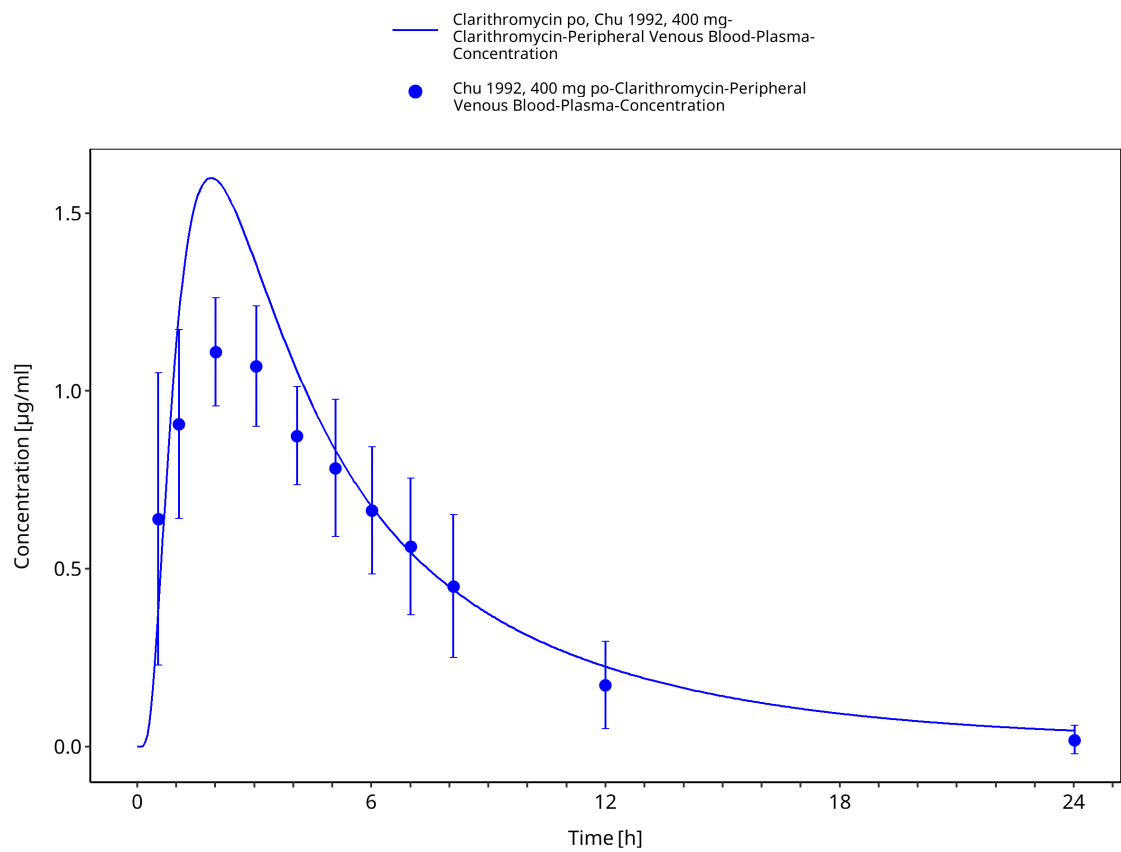


Figure 3-20: Time Profile Analysis 1





**Figure 3-21: Time Profile Analysis**

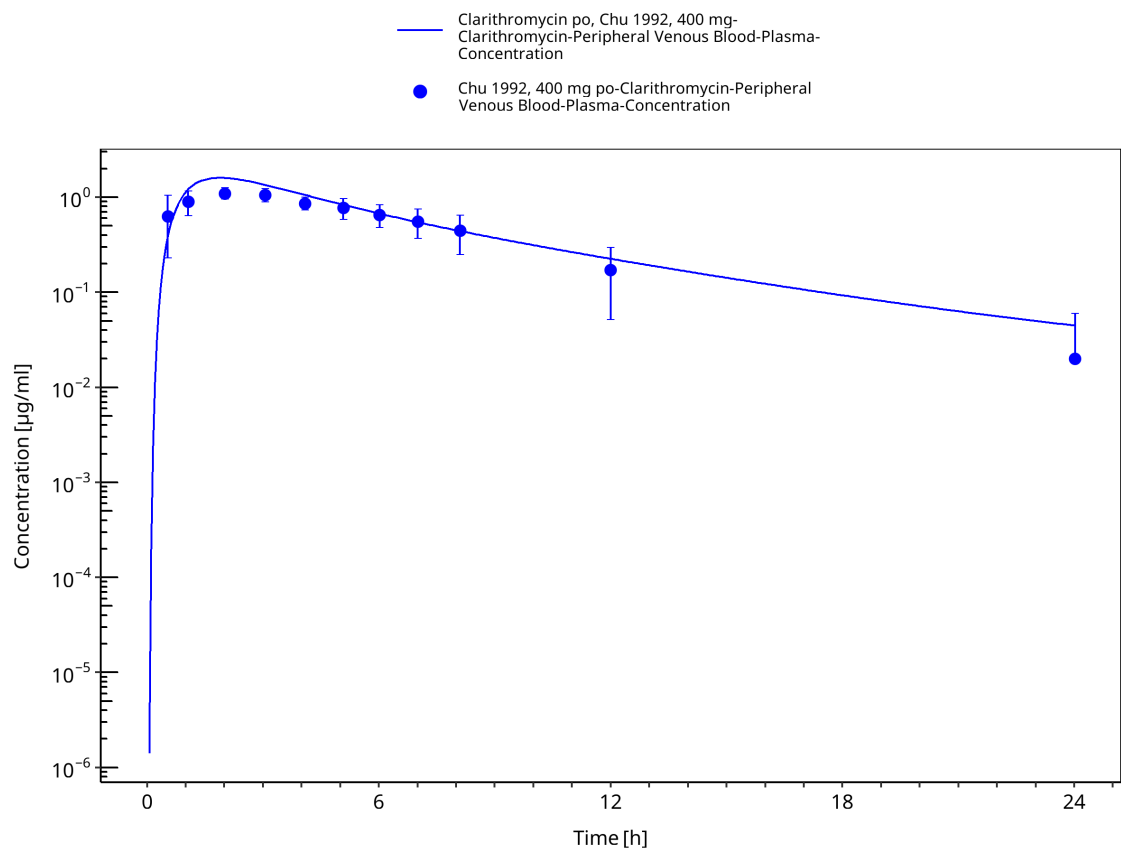
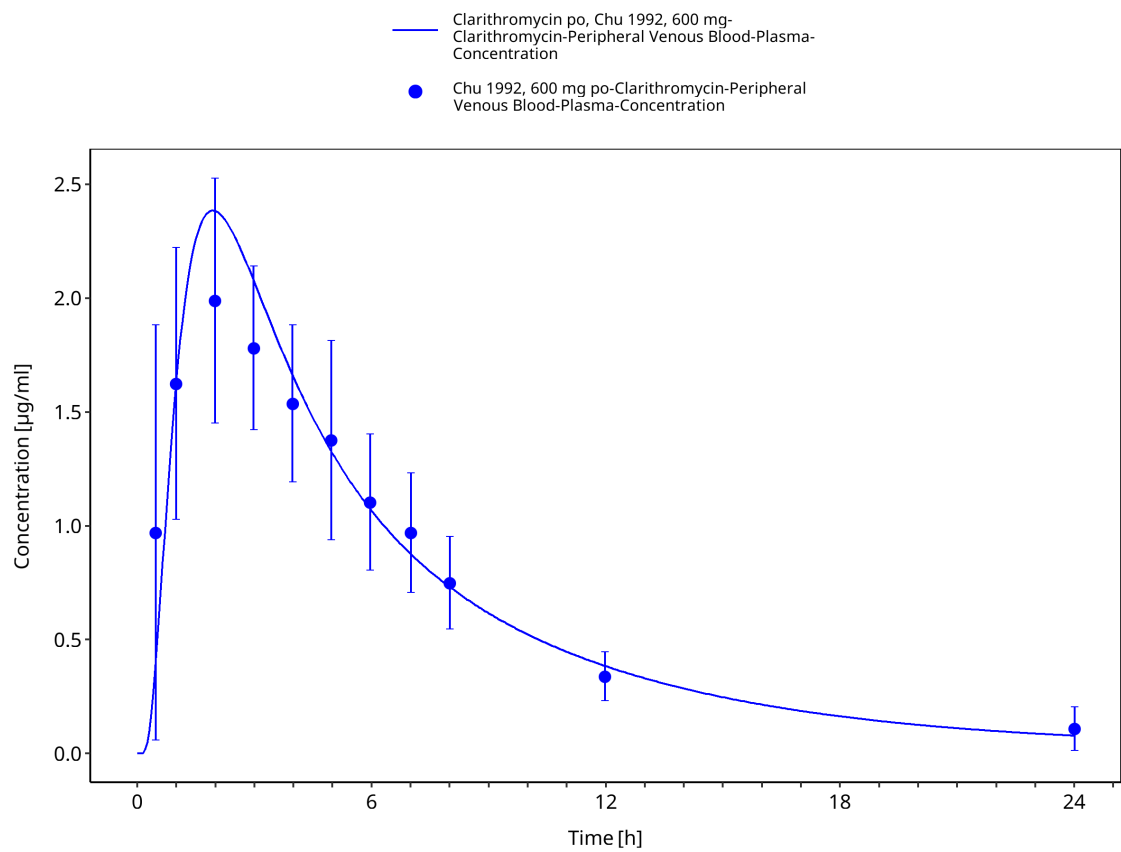


Figure 3-22: Time Profile Analysis 1



**Figure 3-23: Time Profile Analysis**

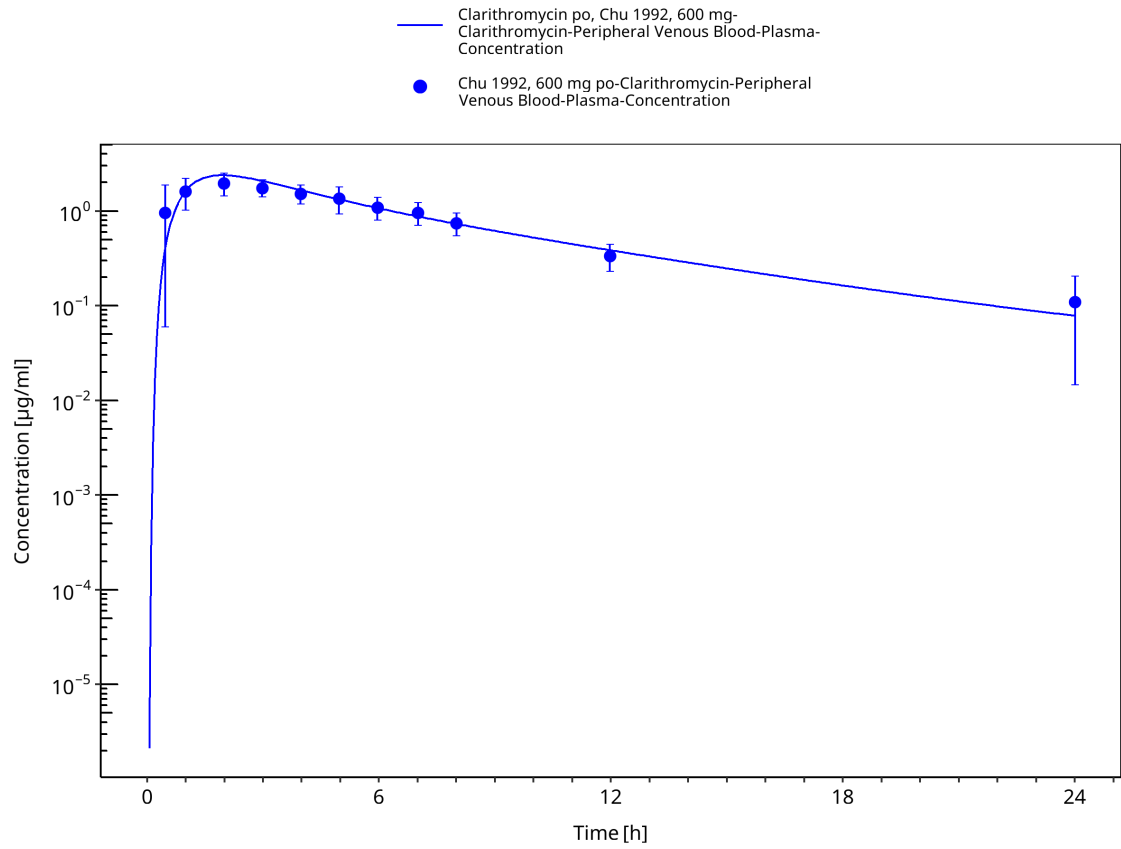
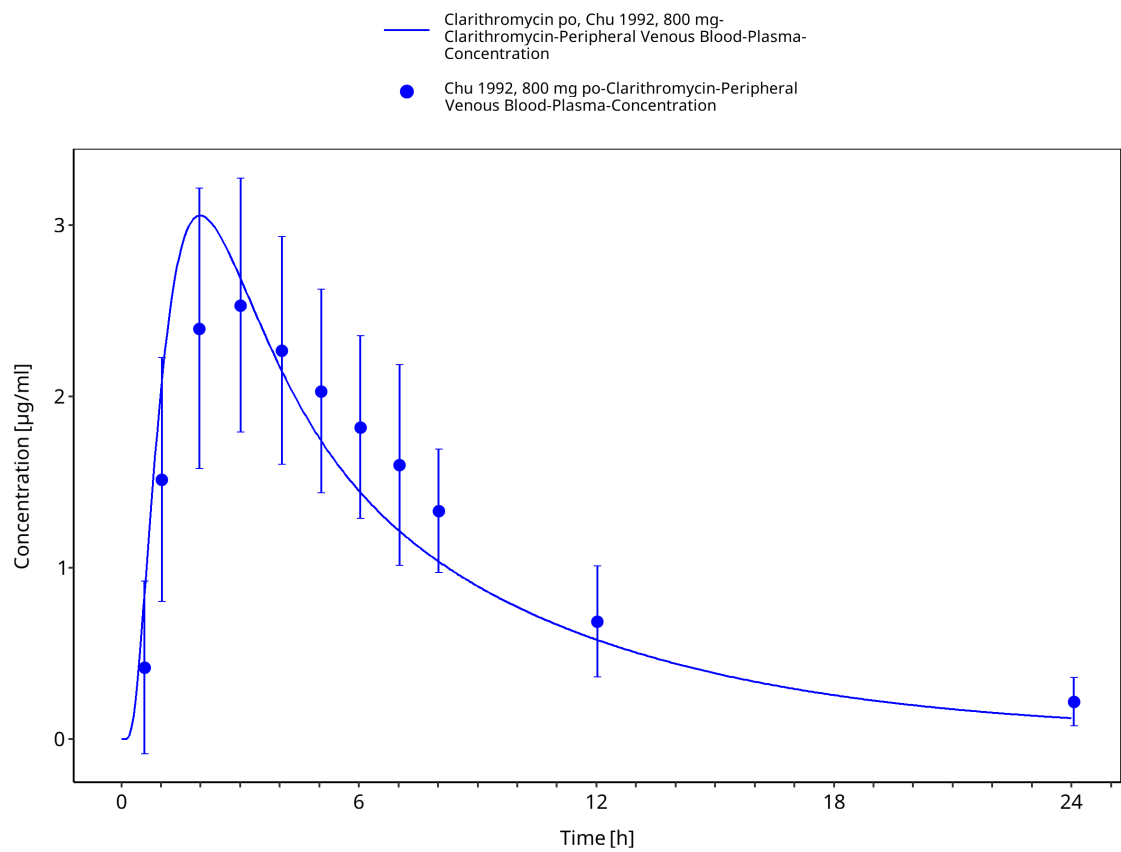


Figure 3-24: Time Profile Analysis 1



**Figure 3-25: Time Profile Analysis**

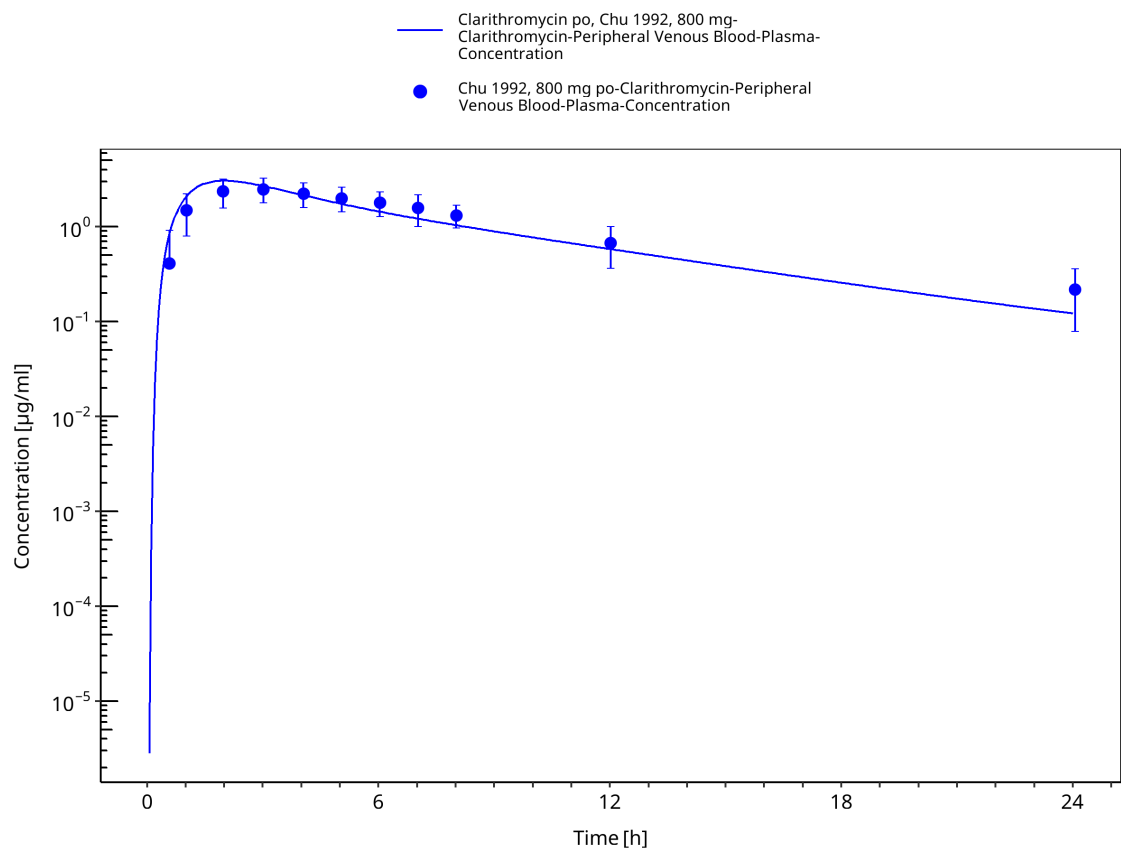
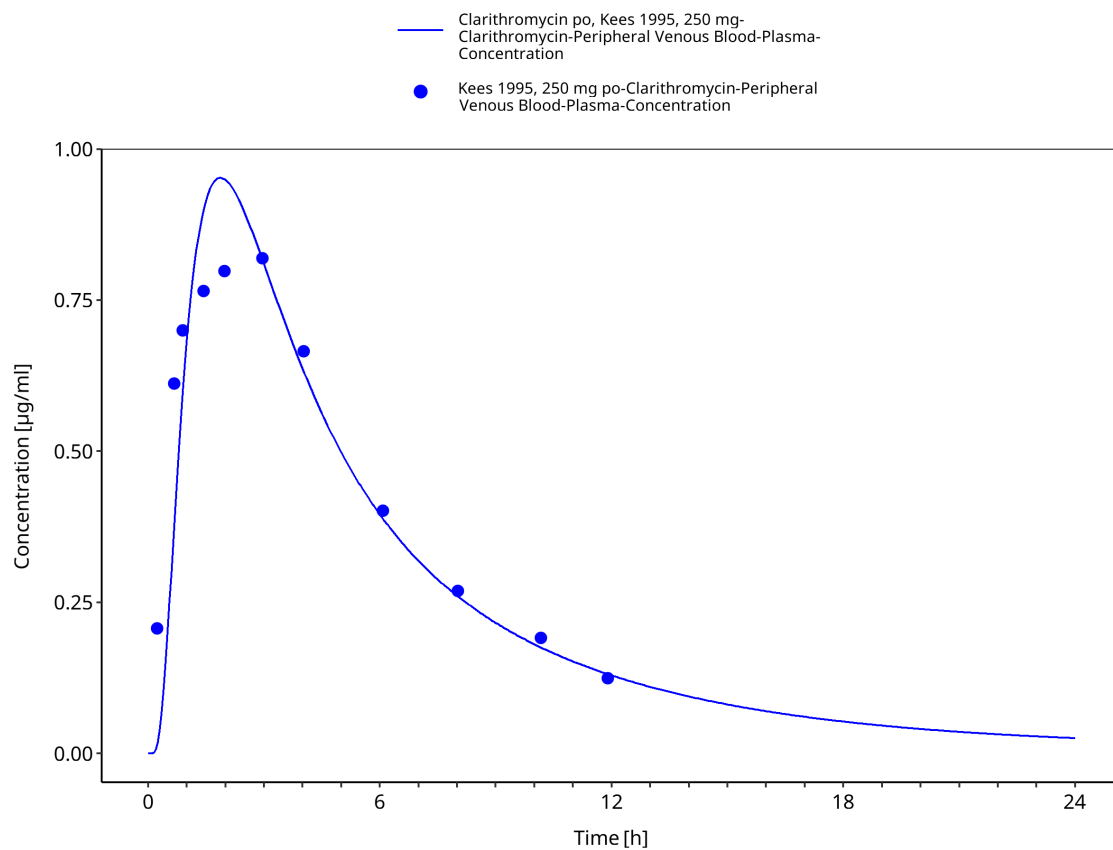
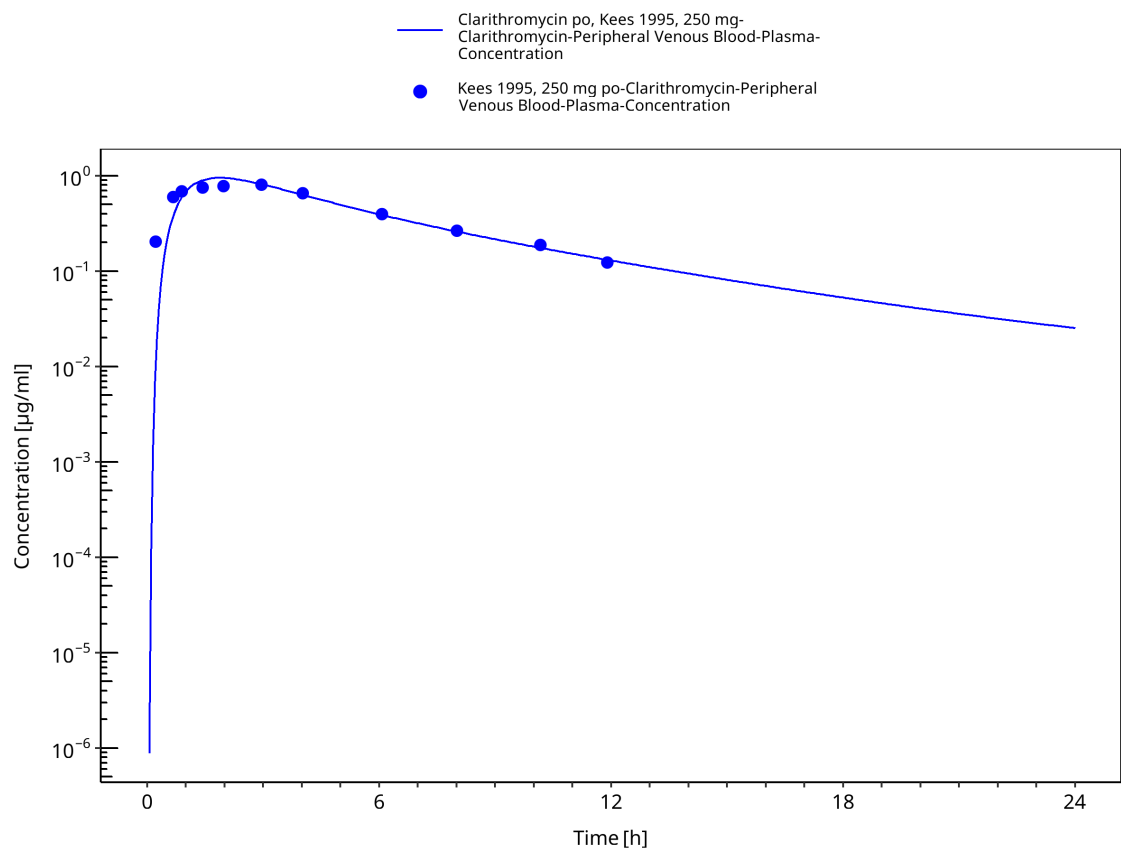


Figure 3-26: Time Profile Analysis 1

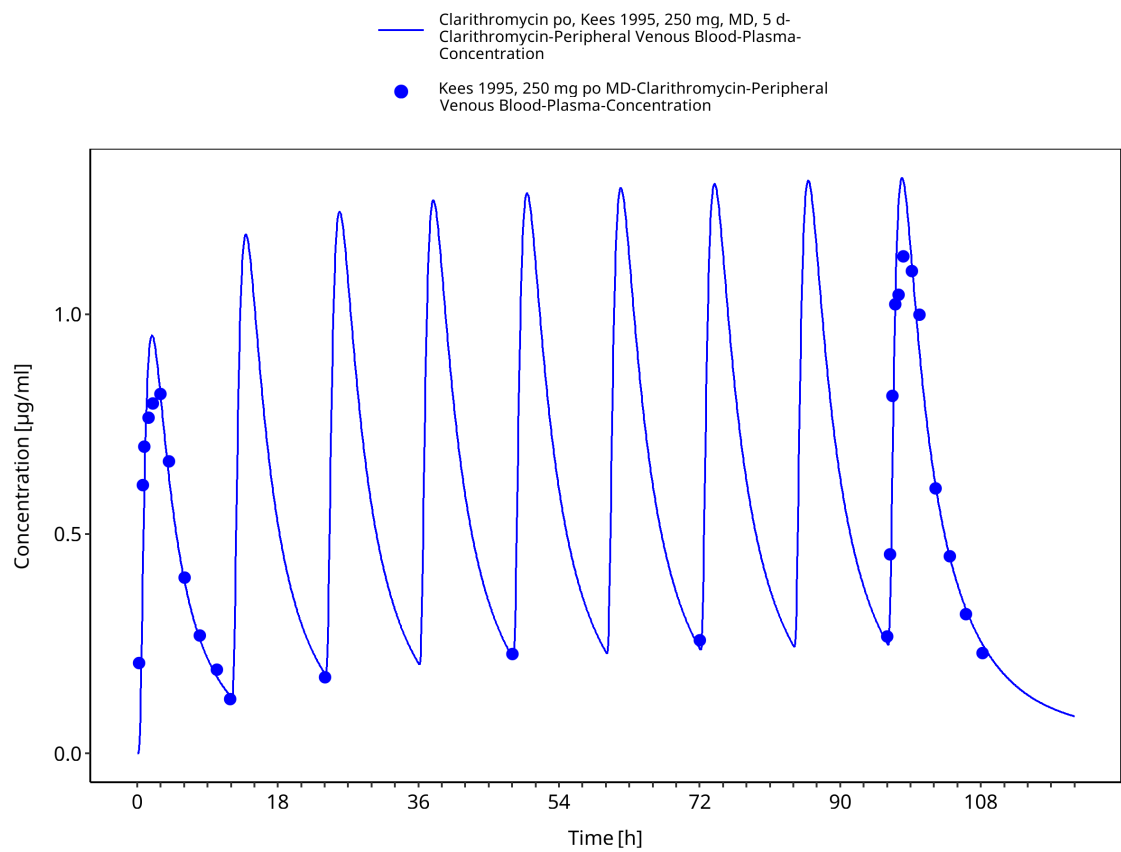


**Figure 3-27: Time Profile Analysis**

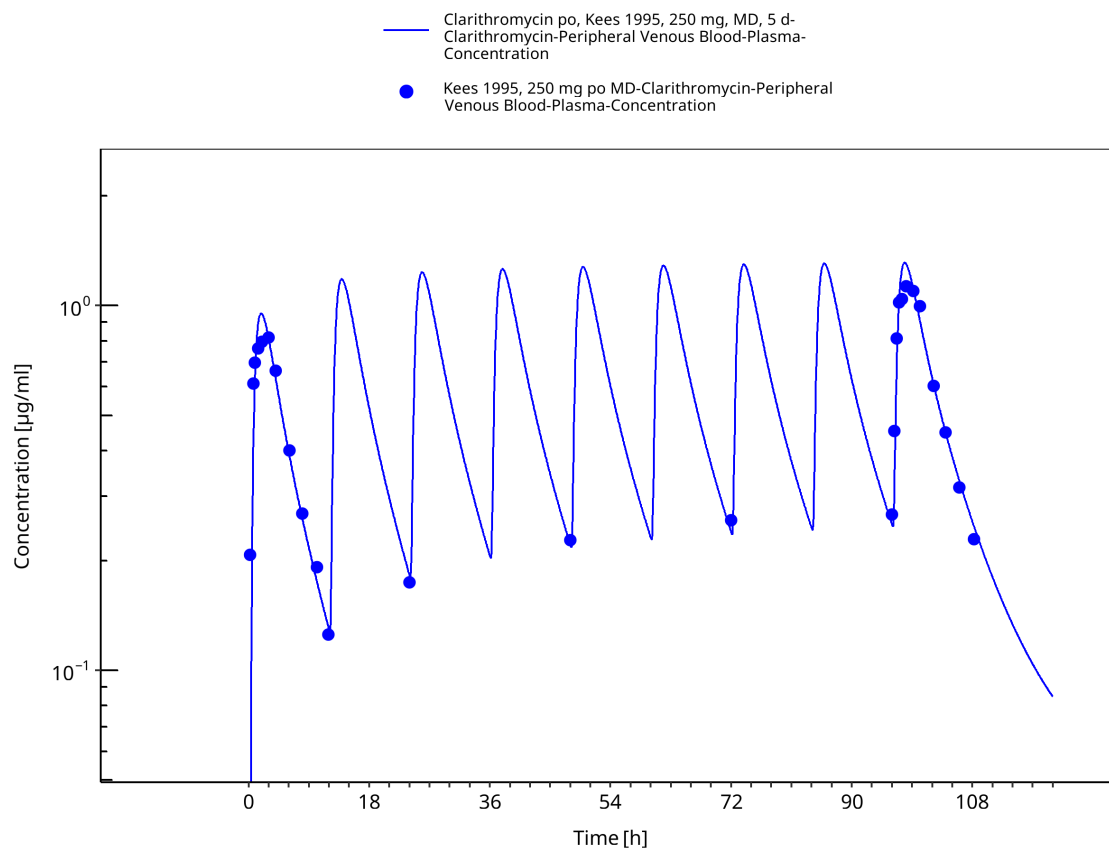


**Figure 3-28: Time Profile Analysis 1**

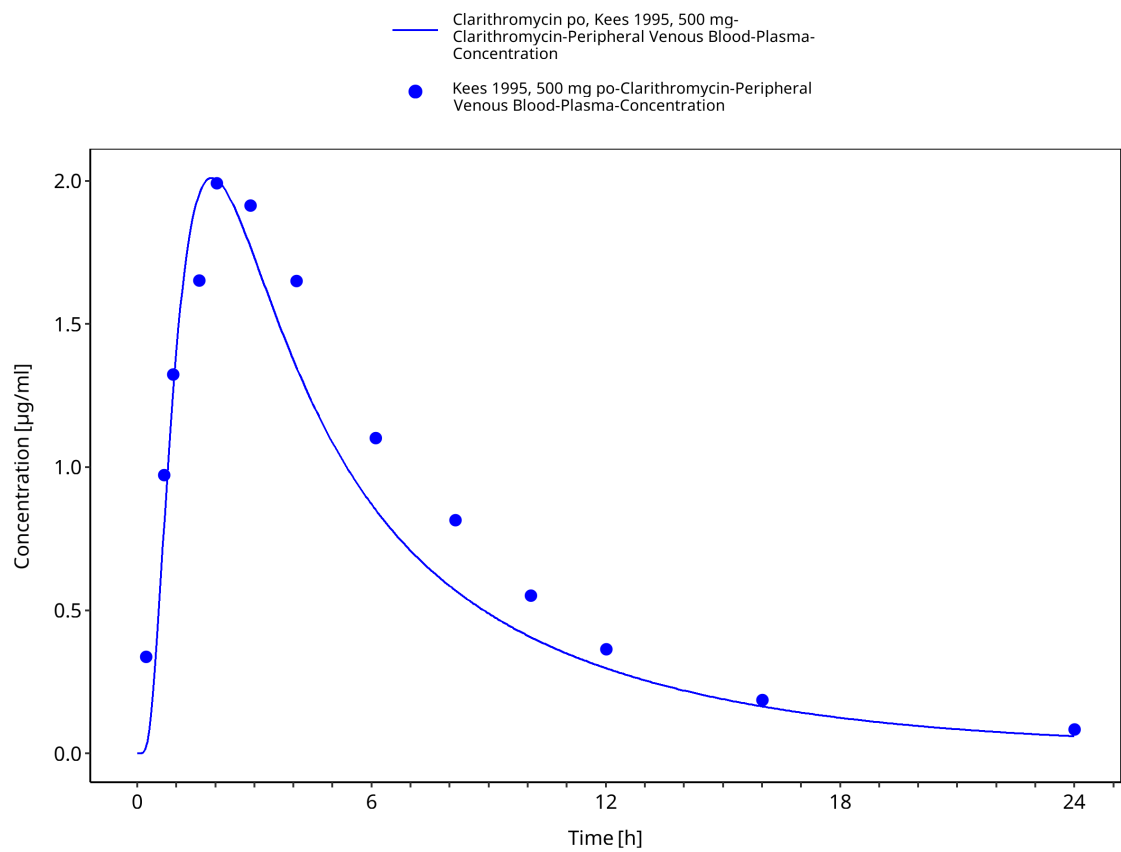




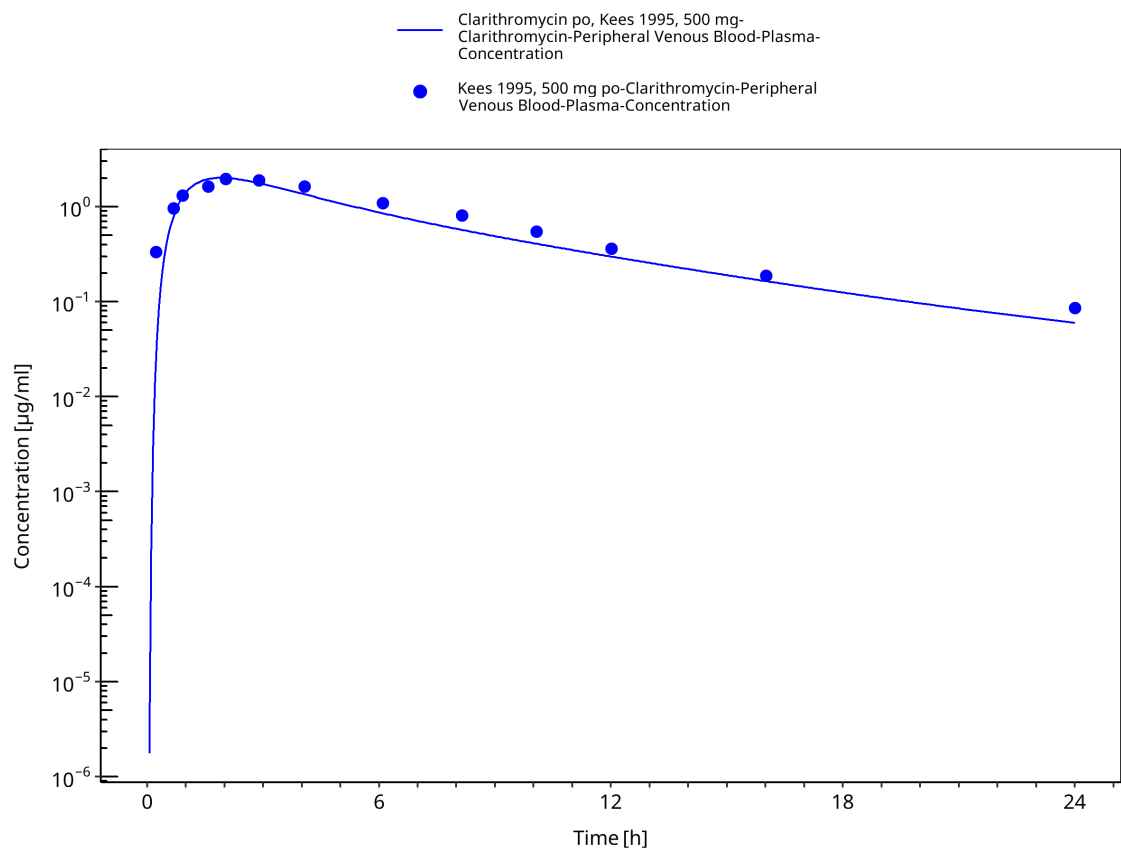
**Figure 3-29: Time Profile Analysis**



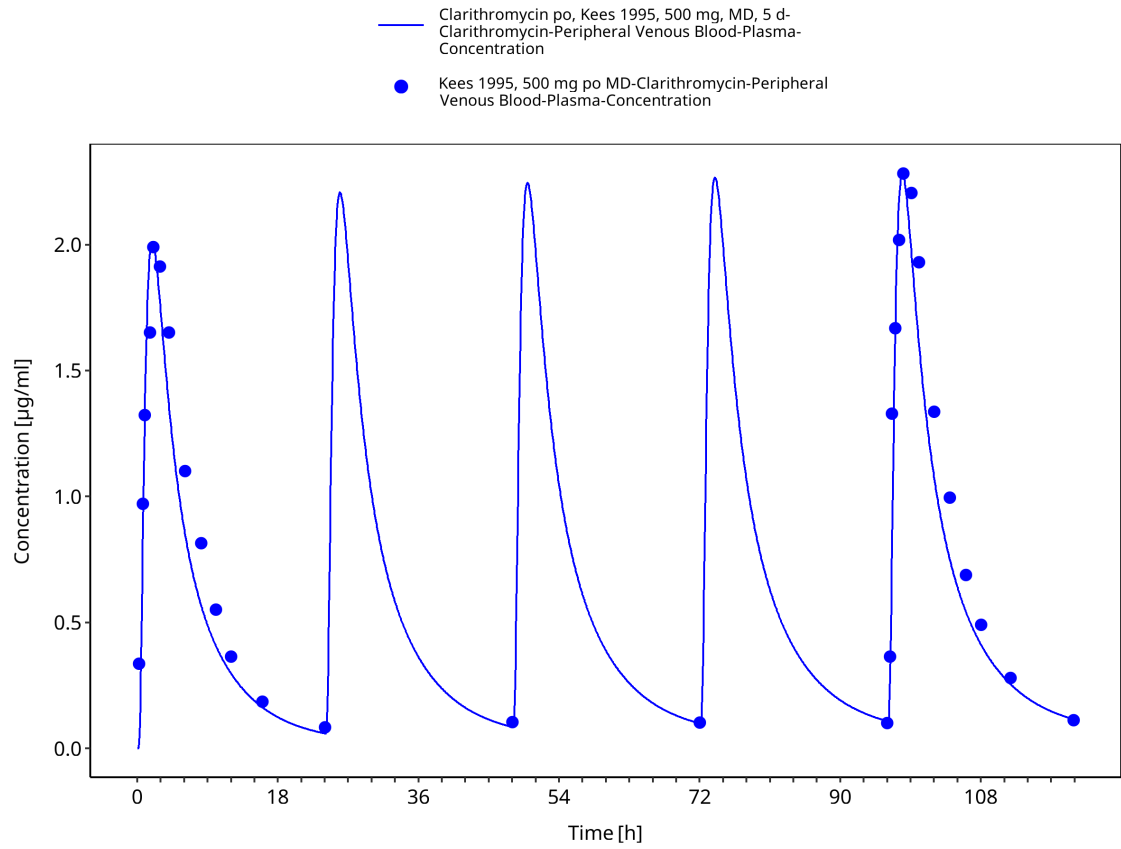
**Figure 3-30: Time Profile Analysis 1**



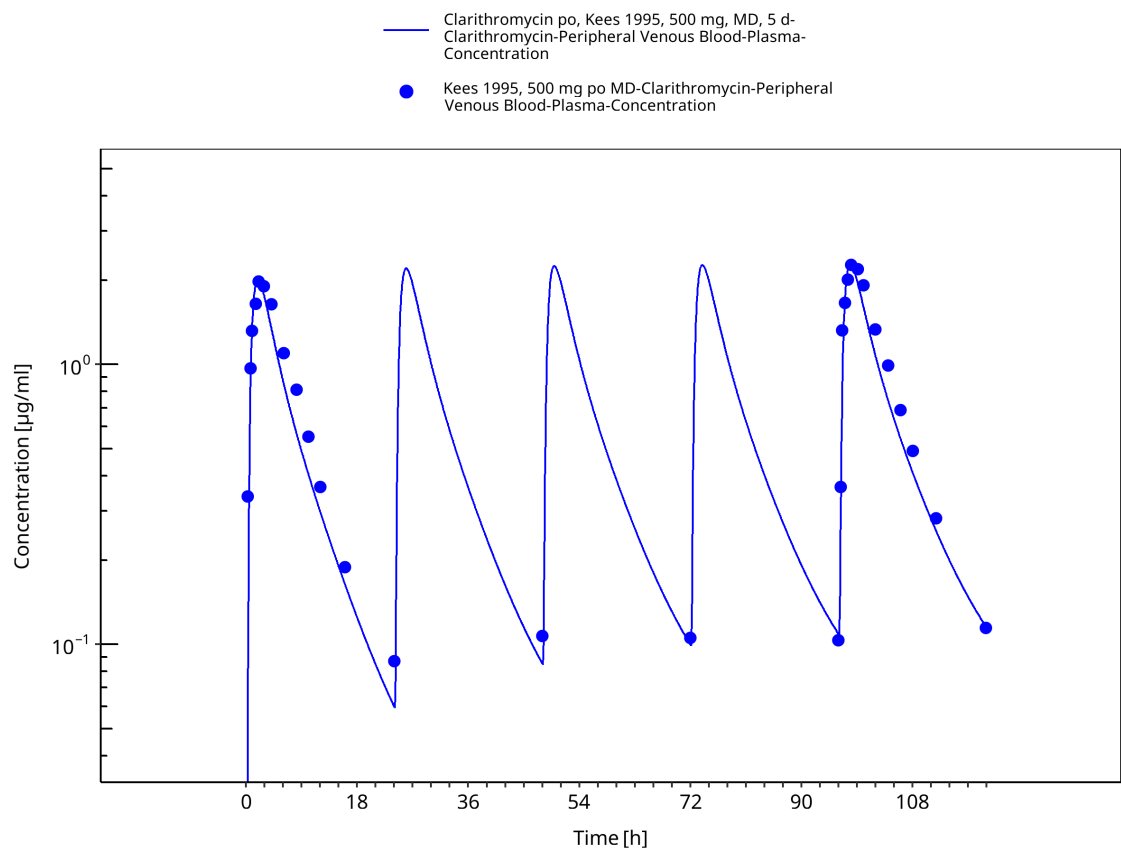
**Figure 3-31: Time Profile Analysis**



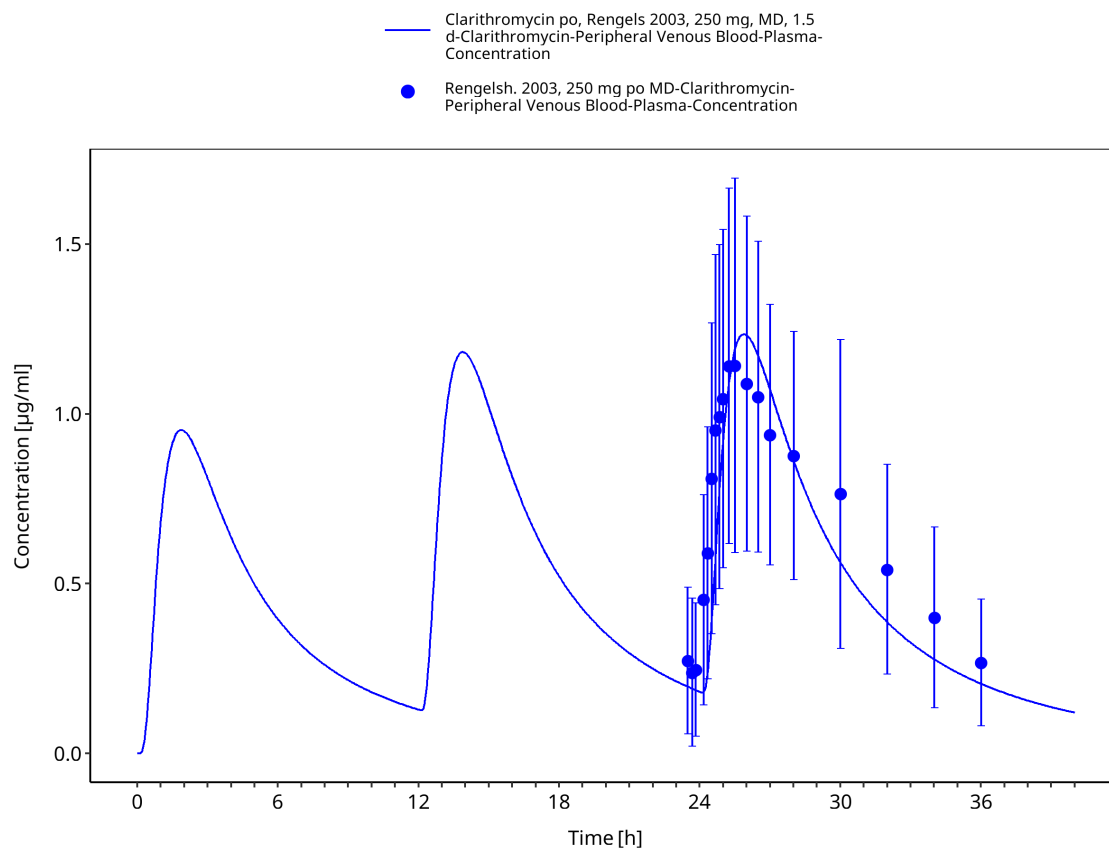
**Figure 3-32: Time Profile Analysis 1**



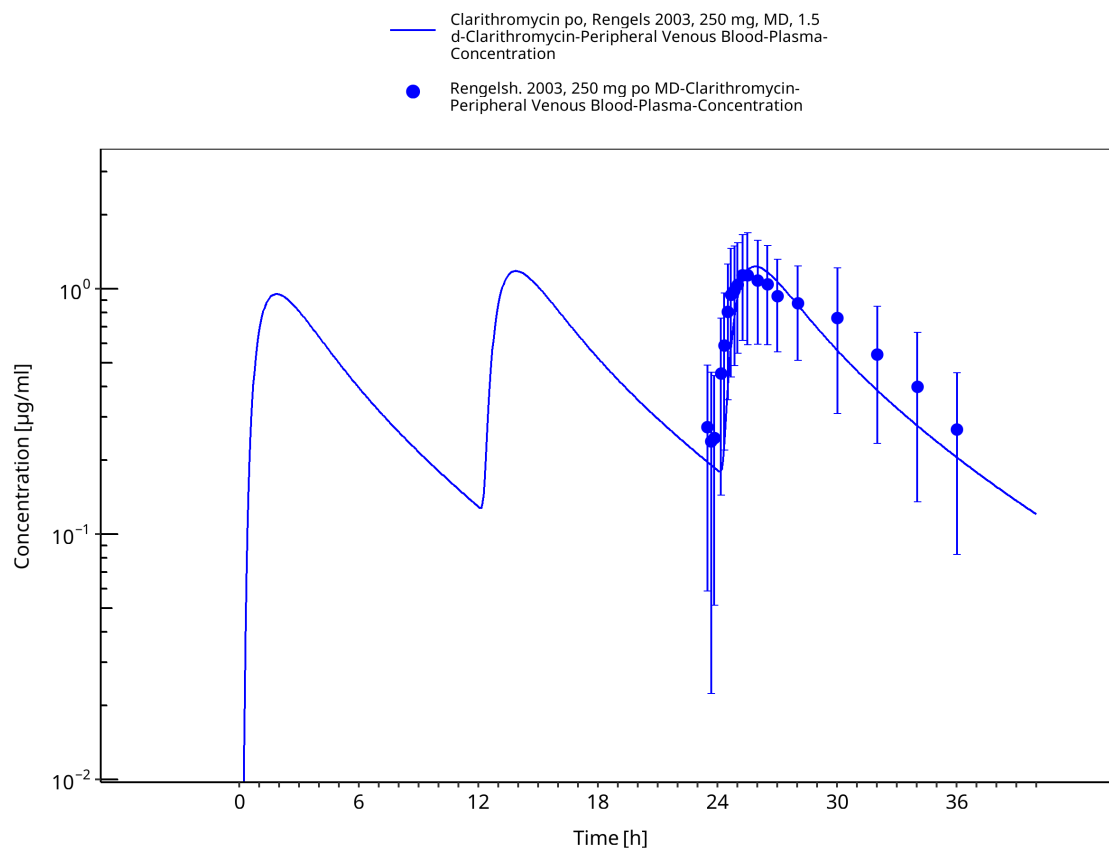
**Figure 3-33: Time Profile Analysis**



**Figure 3-34: Time Profile Analysis 1**



**Figure 3-35: Time Profile Analysis**



**Figure 3-36: Time Profile Analysis 1**



## 4 Conclusion

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The herein presented PBPK model adequately describes the pharmacokinetics of clarithromycin in adults. In particular, it applies increased transfer and accumulation in red blood cells, metabolism by CYP3A4, renal clearance as unchanged drug and mechanism-based inactivation of CYP3A4. Thus, the model is fit for purpose to be applied for the investigation of drug-drug interactions with regard to inhibition of CYP3A4 and P-gp.

## 5 References

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- Abduljalil 2009** Abduljalil, K. et al. Modeling the autoinhibition of clarithromycin metabolism during repeated oral administration. *Antimicrob. Agents Chemother.* 53, 2892–901 (2009).
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